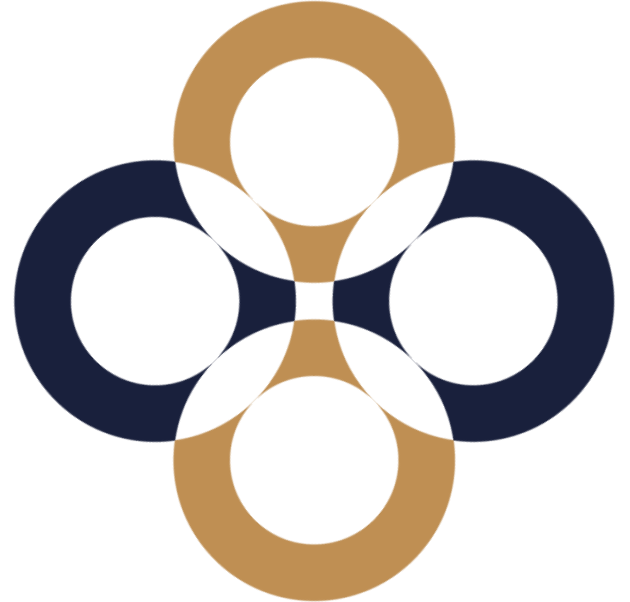
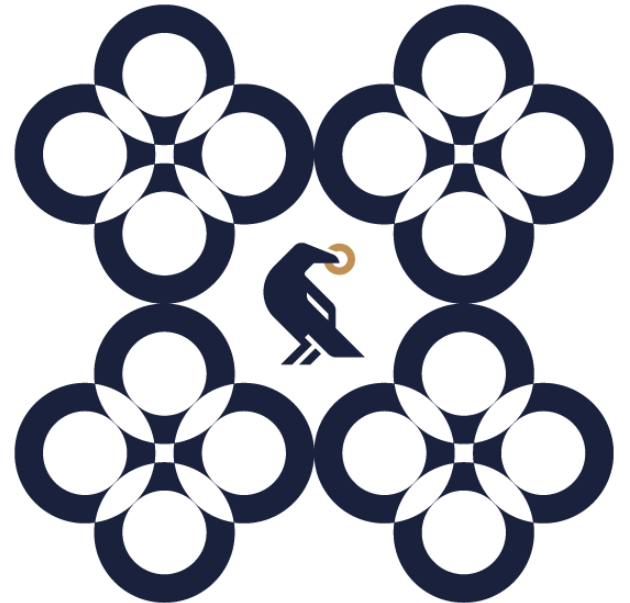


Business valuation specialities of the Financial Service Industry

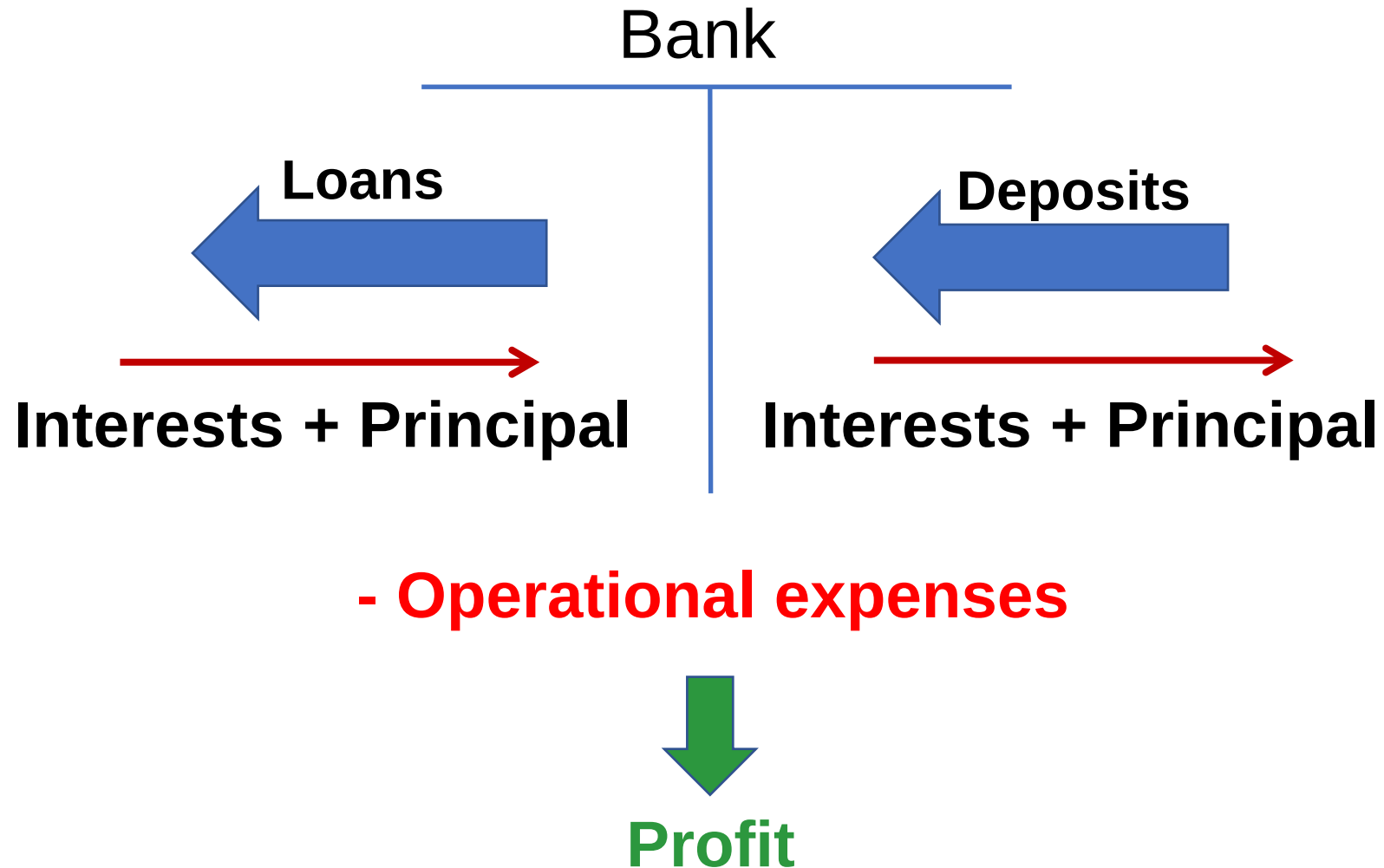
Péter Juhász, PhD, CFA



Commercial & Investment banks



Value creation at banks



Quantity

Lifetime

Risk

Basic business logic

Loans

$$100 * 8\% = 8$$

Deposits

$$100 * 3\% = 3$$

Profit

$$8 - 3 = 5$$

Mistakes:

- Cost of operation?
- Liquidity?
- Loan defaults?

Improved business logic

Assets (100)

Loans

$$90 * 8\% * 95\% = 6.84$$



Default loss

Liquid assets

$$6 * 5\% = 0.30$$

Cash

$$4 * 0\% = 0.00$$

Deposits

$$100 * 3\% = 3.00$$

Cost of operation

$$4.00$$

Assume Equity = 10% * Assets

Profit:

$$6.84 + 0.30 + 0.00 - 3.00 - 4.00 = 0.14$$

Profit/Assets:

$$0.14 / 100 = 0.14\%$$

Profit/Equity:

$$0.14 / 10 = 1.40\%$$

Alternative scenario

Assets

Loans

$$91 * 8.2\% * 98\% = 7.31$$

Liquid assets

$$7 * 5.5\% = 0.39$$

Cash

$$2 * 0.5\% = 0.01$$

Deposits

$$100 * 2.8\% = 2.80$$

Cost of operation

$$3.60$$

Assume Equity = 10% * Assets

Profit:

$$7.31 + 0.39 + 0.01 - 2.80 - 3.60 = 1.31$$

Profit/Assets:

$$1.31 / 100 = 1.31\%$$

Profit/Equity:

$$1.31 / 10 = 13.10\%$$

Types of banks

**Commercial
banks**

Deposits
Loans
Accounts

**Investment
banks**

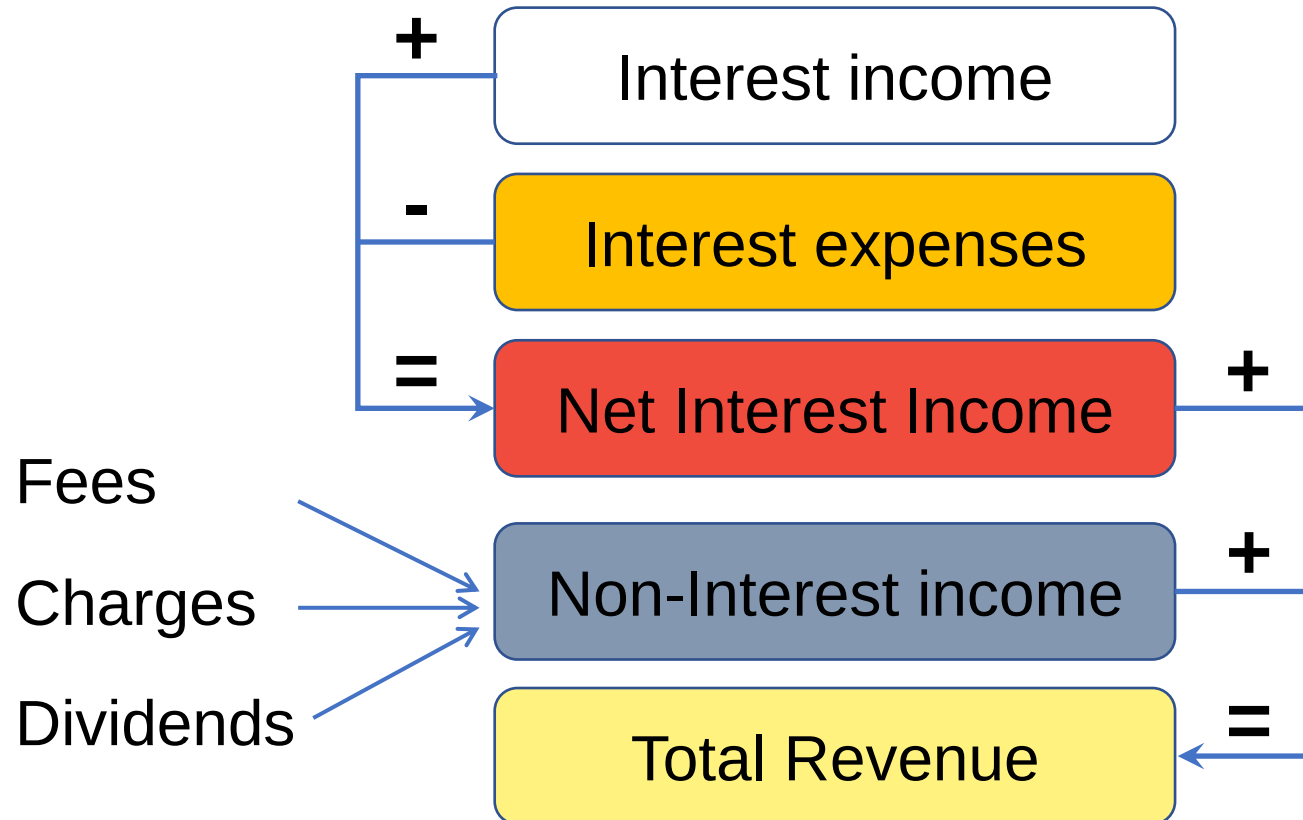
Funds
Trading
Security issue

Universal banks

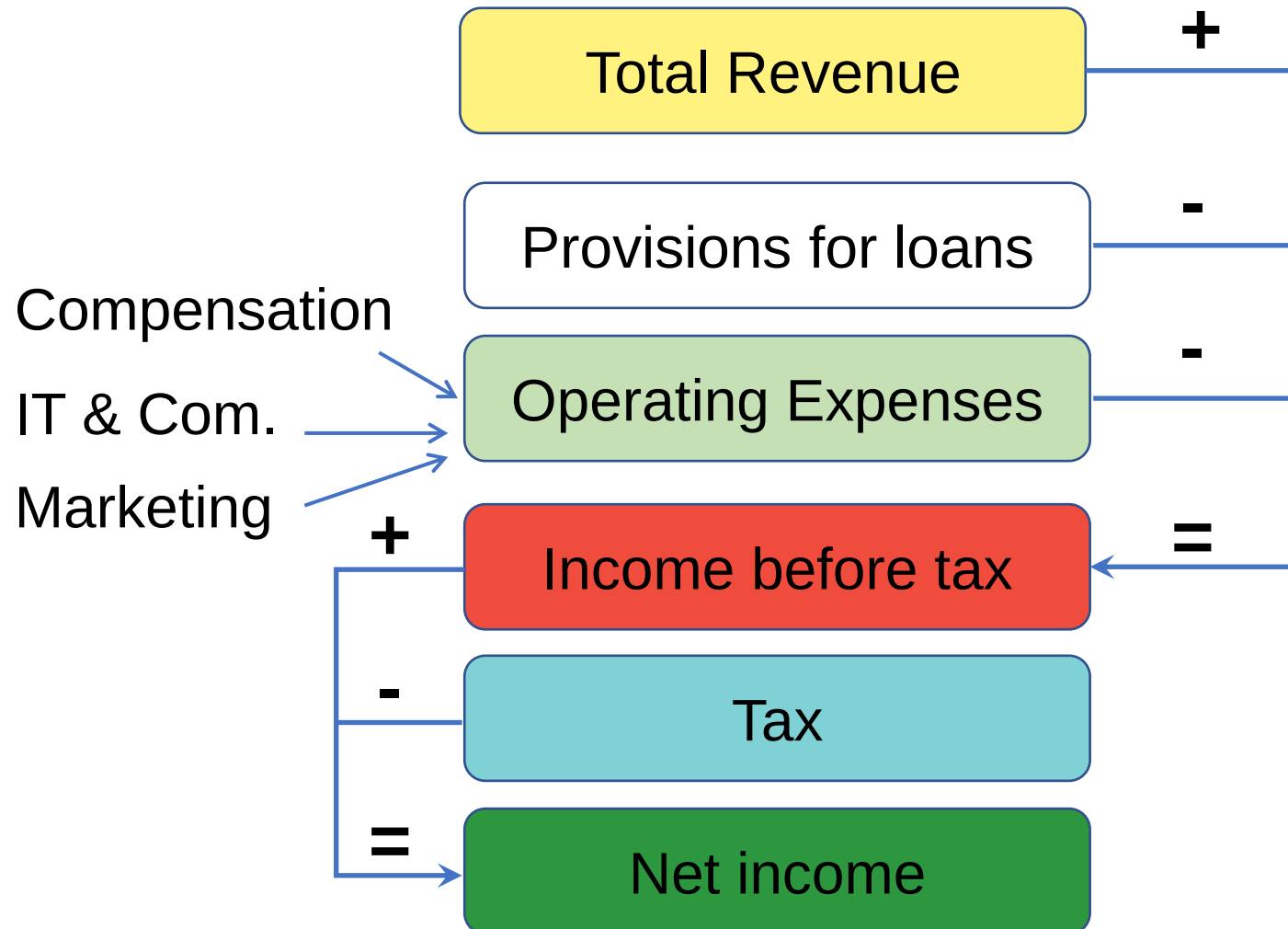
Balance sheet for banks

Cash	Money market deposits
Liquid assets	Time deposits
Loans	CDs, short term loans
Long term investments	Long term debt
Real estate & equipment	Equity

Income statement for banks 1



Income statement for banks 2



Challenges of Analysis

One-time effects
(extraordinary
items)

Change in
strategy / activity
mix

Differences /
changes in
accounting
policy

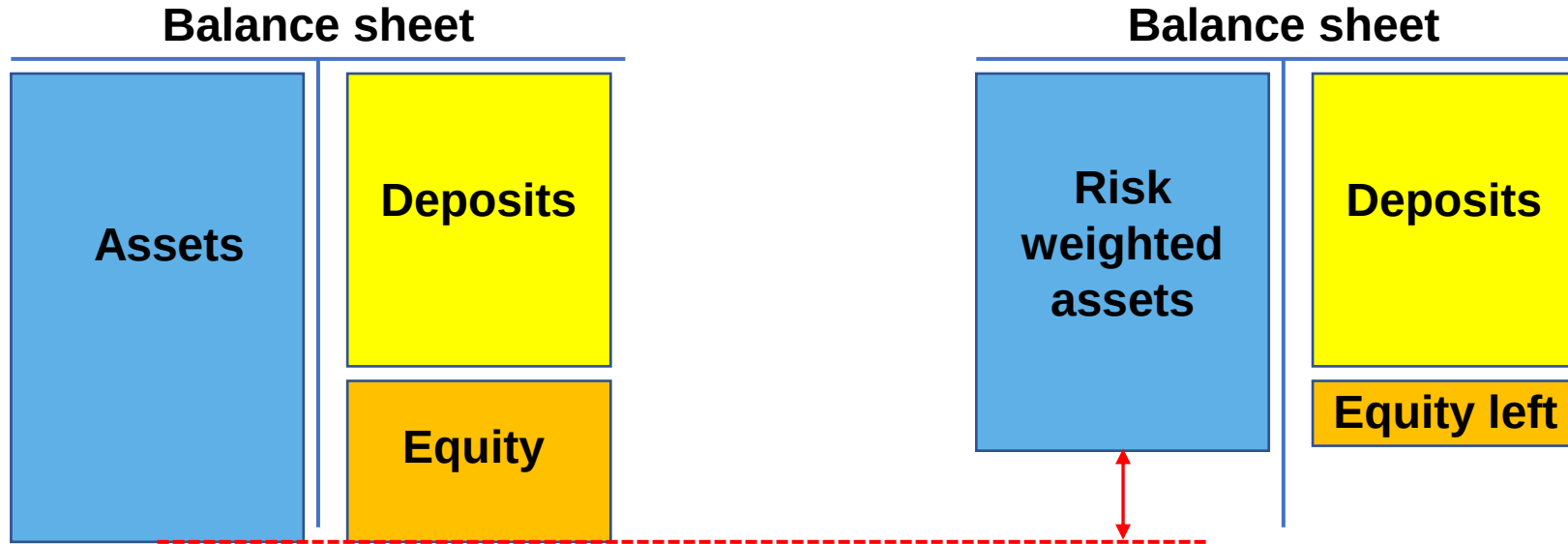
Off-balance
sheet activities

Cosmetics
(Window
dressing)

Capital Adequacy
Asset Quality
Management
Earnings (Profitability)
Liquidity & Funding
Sensitivity to Market Risks



Capital Adequacy 1



Basel III:

BIS Tier 1 Capital: at least 4.5%+1.5% RWA

BIS Tier 2 Capital: at least 2.0% more

Common Equity Tier (CET) 1:
Common stock (equity shares)
Retained earnings
Other comprehensive income
Certain reserves

Additional T1:
Perpetual preferred shares
Instruments with no fixed
maturity or repayment obligation

Tier 2:
Subordinated Debt
Hybrid Instruments
Loan-Loss Reserves
Revaluation Reserves

Basel / U.S. banking regulation: „Well capitalised”:

Leverage Ratio > 5.0%

$$\frac{\textit{Tier 1 capital}}{\textit{Average total assets}}$$

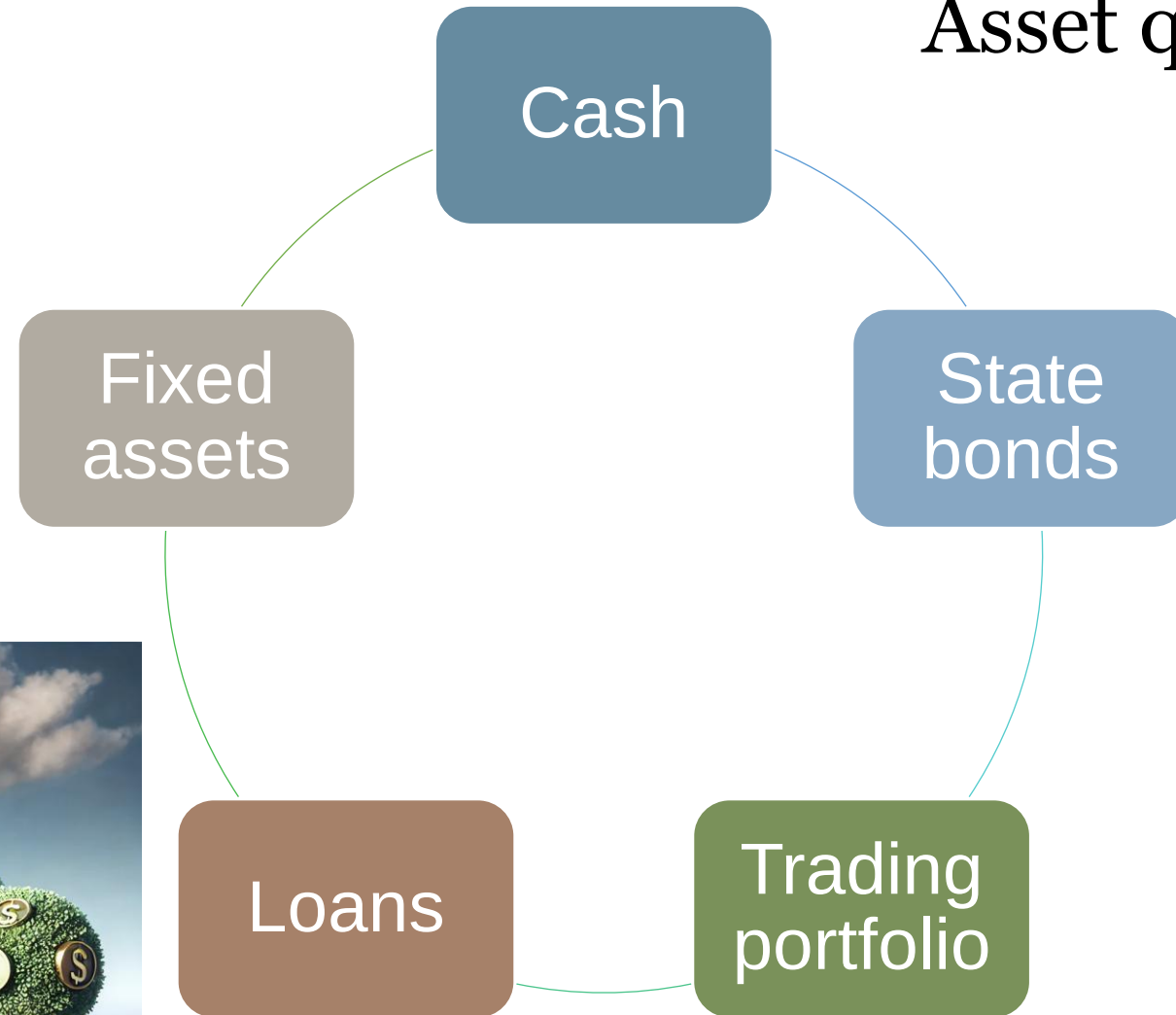
CET1 Capital Ratio > 6.5%

$$\frac{\textit{Tier 1 capital}}{\textit{Total risk adjusted assets}}$$

Total Capital Ratio > 10.0%

$$\frac{\textit{Total capital}}{\textit{Total risk adjusted assets}}$$

Asset quality 1



Loan structure:

Consumer vs Commercial

Term Structure

Industry Exposure

Loan loss reserve

Asset growth rate



How to measure asset quality?

Loan loss reserve / Total loans

Coverage ratio (> 1.5)

Loan Loss Reserves

Non-Performing or Non-current Loans and leases

Overdue Loans to Total Loan Ratio (30-89 days)

90-Day Overdue Loans to Total Loans Ratio (90 days)

Exposure Structure

%	Total Exp.	Uncovered (Risk)	Uncov./ Total exp.
Agriculture	0.01	0.03	52.14
Oil, Gas, Mining	1.61	1.51	17.39
Manufacturing	3.82	6.99	33.74
Real Estate & Construction	19.50	36.29	34.30
Trade	7.47	16.27	40.16
Transp., Storage & Com.	0.41	0.01	0.50
Financial Institutions	16.07	11.91	13.67
Services	11.36	13.58	22.05
Government	16.52	12.38	13.82
Banking	21.19	0.06	0.05
Other	3.81	0.09	0.44
Total	100.00	100.00	18.44

Privatised

Management

Governance

Owners

Management

Branches



Earnings & Profitability

Strategic
focus

Macro
situation

Regional
structure

Interest vs
non-interest
income

Growth
trends

Extraordinary
events

How to increase interest income?

Provide loan to more risky customers
(quantity)

Provide more loan to riskier customers
(structure)

How to measure profitability? 1

Net interest margin

$$\frac{\text{Net Interest Income}}{\text{Average Interest Earning Assets}}$$

Return on (Average) Assets (ROAA)

(US: 0.6-2.0%)

$$\frac{\text{Net Operating Income}}{\text{Total (Average) Assets}}$$

Return on (Average) Equity (RO_{AE})

(US: 15-17%)

$$\frac{\text{Net Income After Tax}}{\text{Total (Average) Equity}}$$

How to measure profitability? 2

Return on Earning Assets (RO_{EA})

$$\frac{\text{Revenues Before Interest Expense}}{\text{Earning Assets}}$$

Operating Profit Margin

$$\frac{\text{Operating Profits}}{\text{Net operating revenues}}$$

Overhead ratio

$$\frac{\text{Non – Interest Expenses}}{\text{Total (Average) Assets}}$$

How to measure profitability? 3

Average Collection Days of Interests

$$\frac{\text{Accured Interest Receivables}}{\text{Interest income}} * 365$$

Efficiency Ratio

(below 40% very efficient, above 75% very expensive)

$$\frac{\text{Non – interest Expenses}}{\text{Total Income (Net int. + Non – int.)}}$$

Du-Pont analysis for banks

$$\begin{aligned}
 ROE &= \frac{\text{Net Income}}{\text{Equity}} = \underbrace{\left(\frac{\text{Total Assets}}{\text{Equity}} * \frac{\text{Total Revenue}}{\text{Total Assets}} * \frac{\text{Income Before Tax}}{\text{Total Revenue}} * \frac{\text{Net Income}}{\text{Income Before Tax}} \right)}_{\text{Net profit margin}} \\
 &\quad \underbrace{\left(\frac{\text{Total Assets}}{\text{Equity}} * \frac{\text{Total Revenue}}{\text{Total Assets}} \right)}_{\text{Equity multiplier} * \text{Asset Utilisation}} \quad \underbrace{\left(\frac{\text{Income Before Tax}}{\text{Total Revenue}} * \frac{\text{Net Income}}{\text{Income Before Tax}} \right)}_{\text{Gross Profit Margin} * \text{Tax burden}} \\
 &\quad \underbrace{\left(\frac{\text{Total Assets}}{\text{Equity}} * \frac{\text{Total Revenue}}{\text{Total Assets}} \right)}_{\text{ROA}} \quad \underbrace{\left(\frac{\text{Income Before Tax}}{\text{Total Revenue}} * \frac{\text{Net Income}}{\text{Income Before Tax}} \right)}_{\text{Tax burden}} \\
 &\quad \underbrace{\left(\frac{\text{Interest Inc.}}{\text{Assets}} + \frac{\text{Non - Int. Inc.}}{\text{Assets}} + \frac{\text{Gain on Sec.}}{\text{Assets}} \right)}_{\text{ROA}}
 \end{aligned}$$

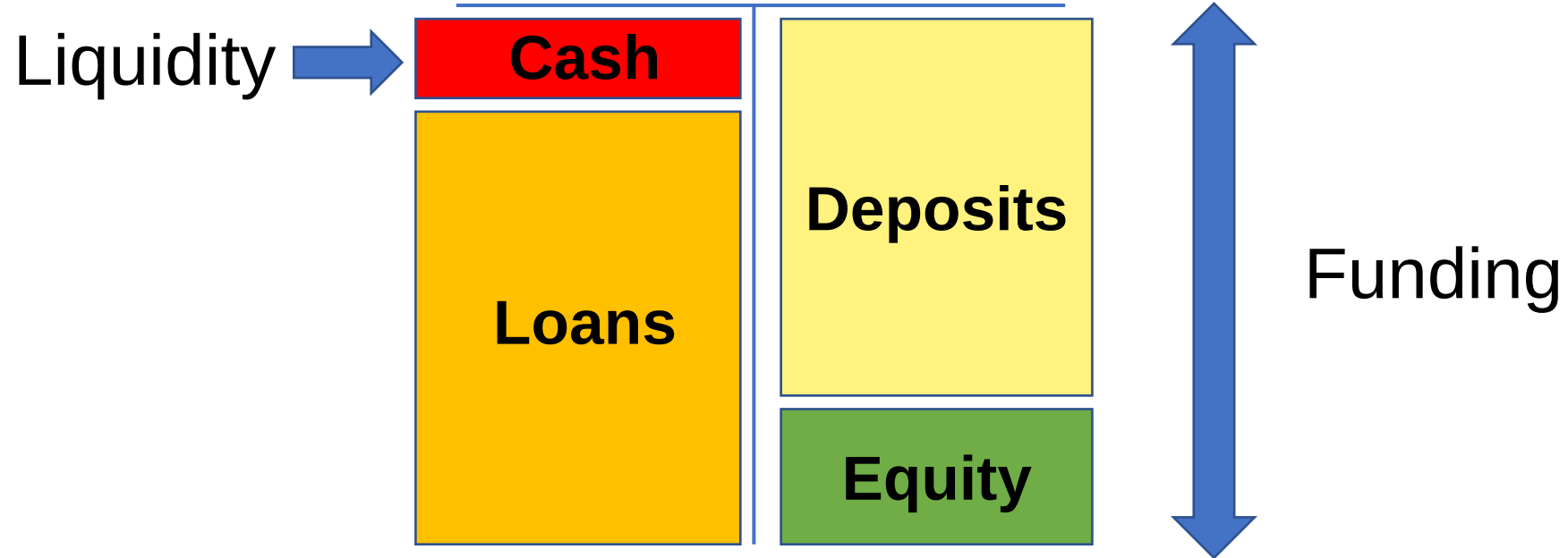


Which bank is better?

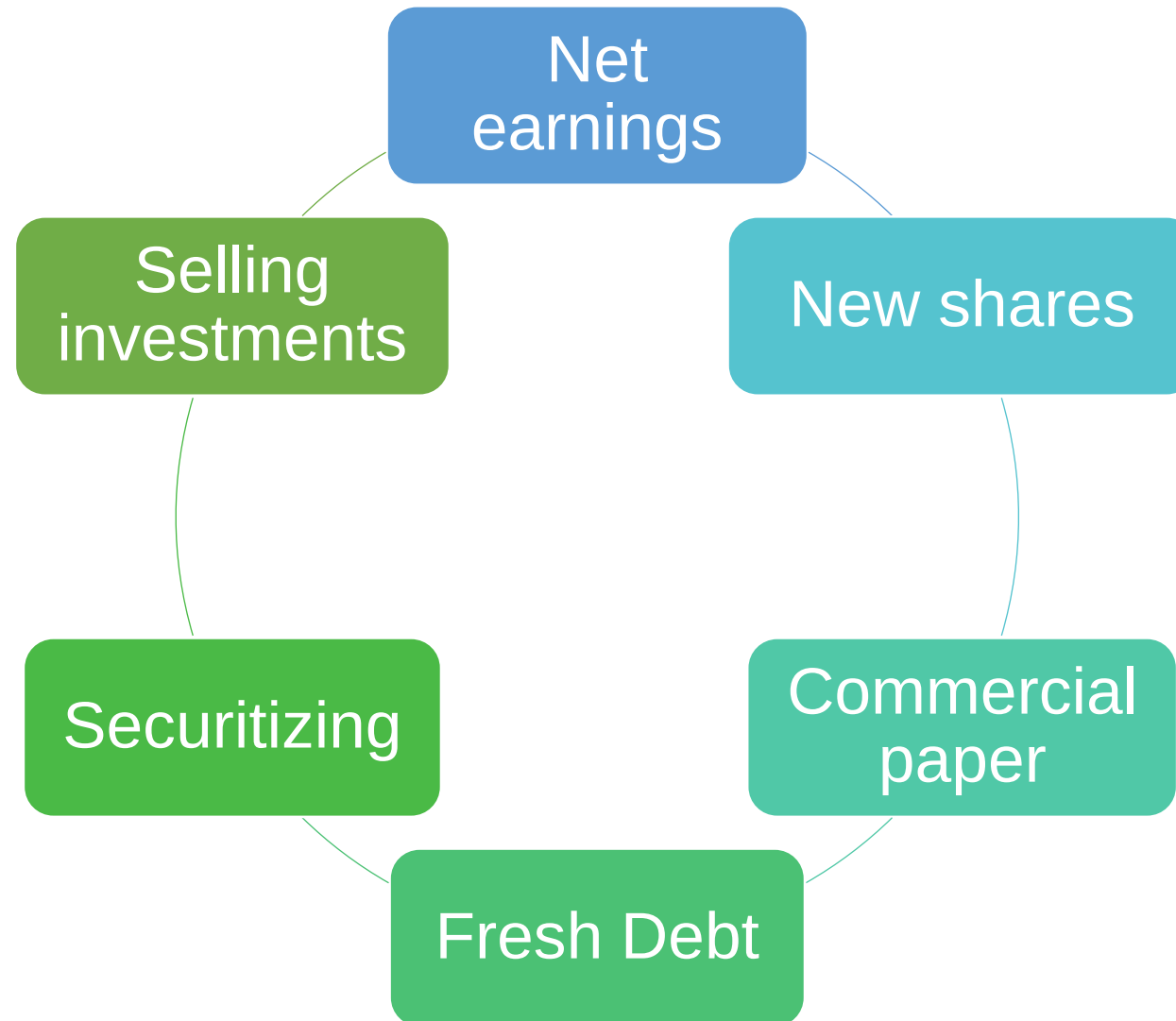
	A	B	C
ROAE	13.75%	12.38%	14.40%
Equity multiplier	5.00	5.50	6.00
Asset utilisation	5.00%	4.50%	4.00%
Net profit margin	55.00%	50.00%	60.00%

More profitable $\neq$ Better

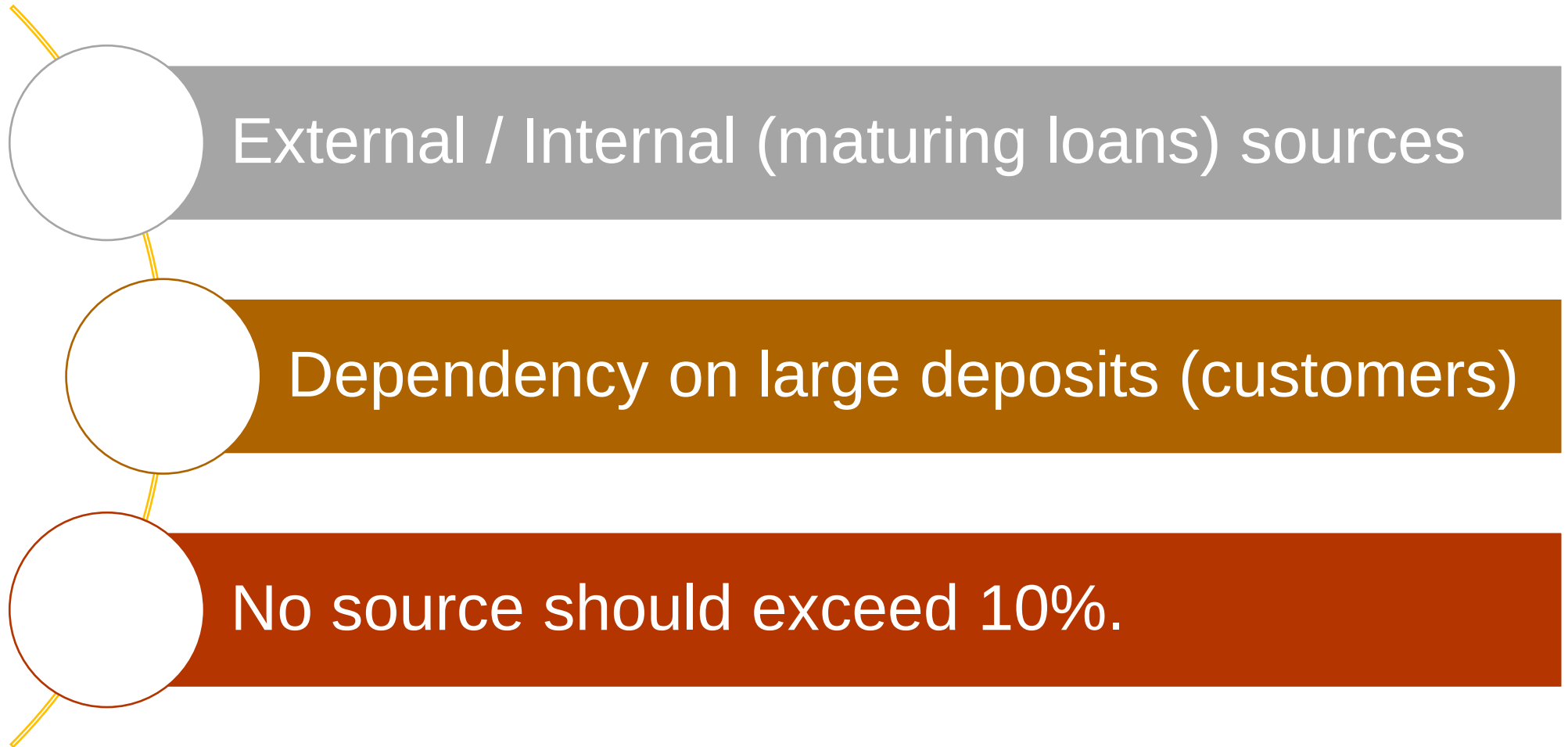
Liquidity & Funding



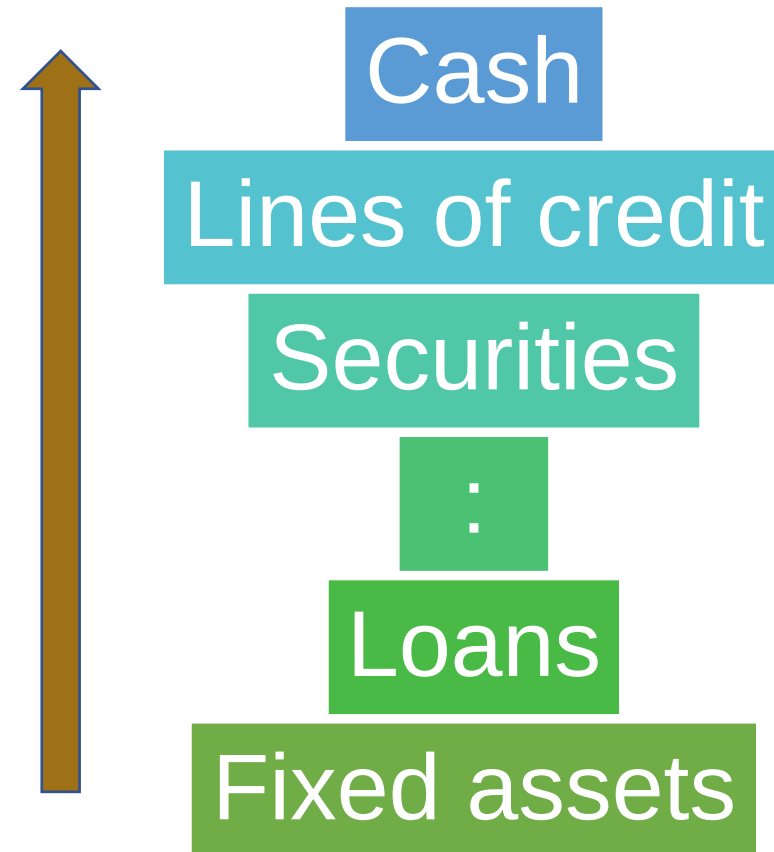
Sources of funding



To check for funding...



...is the bank's ability
to convert an asset
into cash.
... is a question of time.



Predicting direct cash flow:

- + Assets maturing (principal payments)
- Assets increasing (loan demand, investment)
- Liabilities maturing (deposit withdrawals)
- + Liabilities increasing (new deposits, fresh capital)

Sensitivity to Market (Risk)

Interest rate

Operation

Credit

Funding

Liquidity

Market

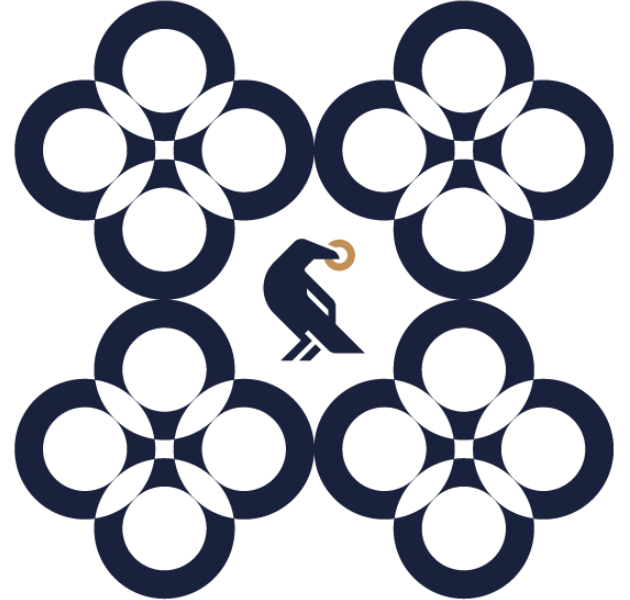
Country

Strategic

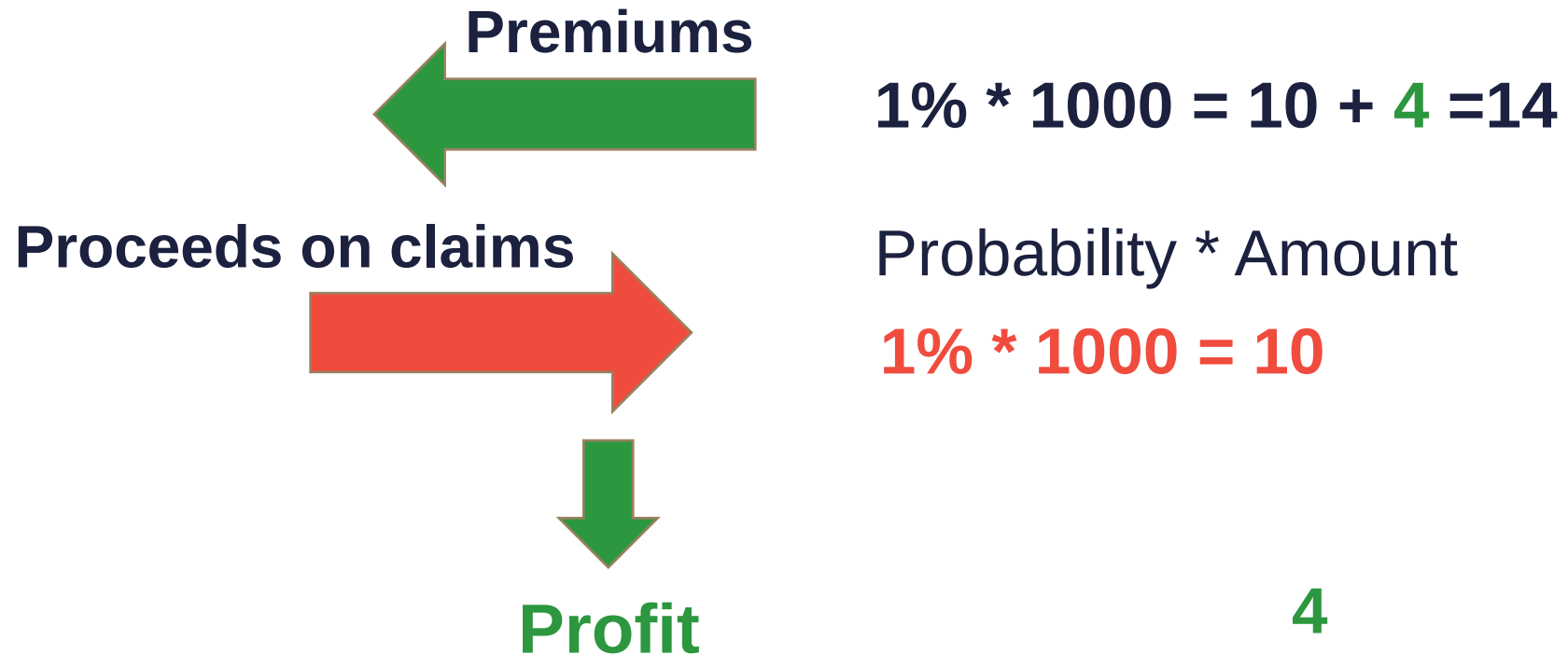
Compliance

Reputation

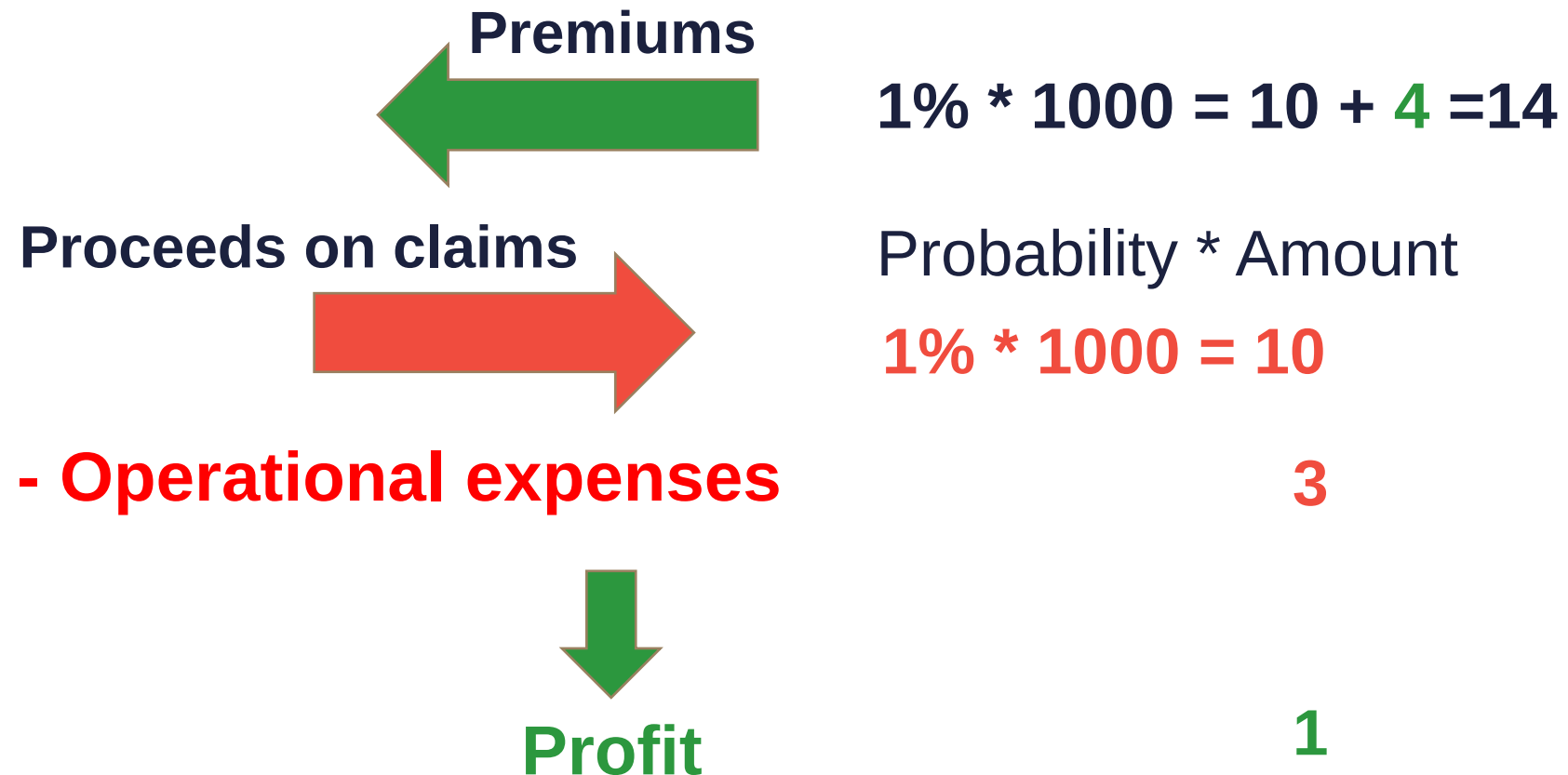
Property and Casualty Insurance



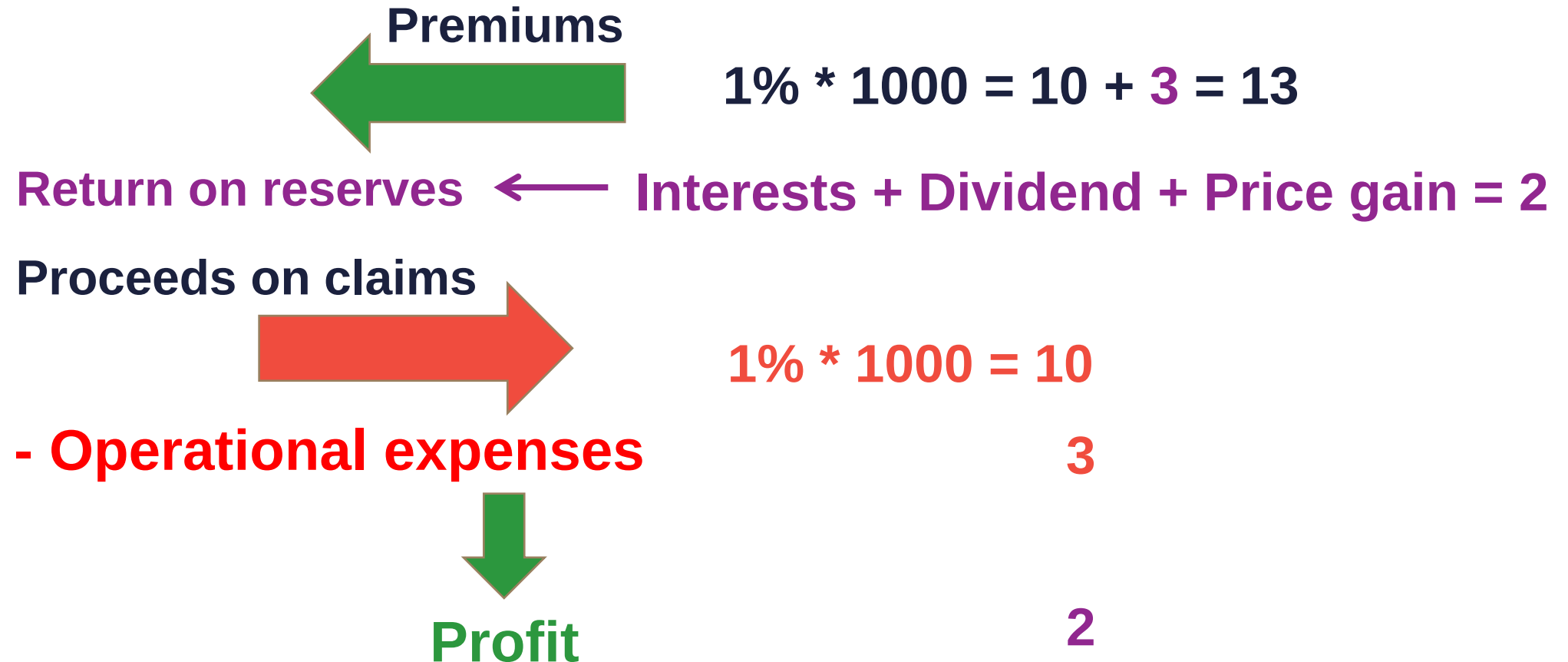
How do P&C Insurance firm function? 1.



How do P&C Insurance firm function? 2.



How do P&C Insurance firm function? 3.



Balance sheet of insurance companies

**Cash and cash like
items**

Receivables

**Reinsurance
assets**

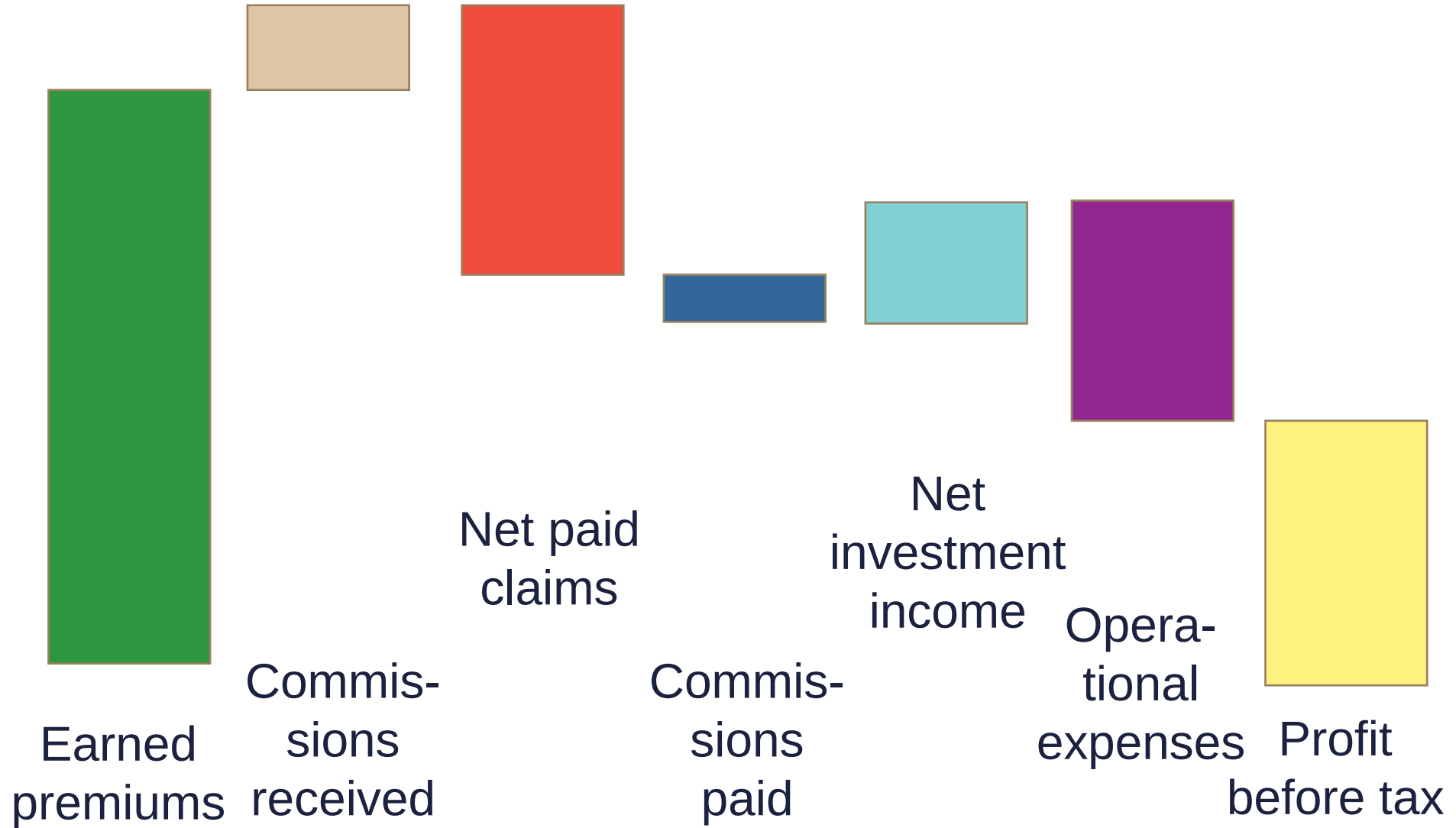
Investments

**Plant, Property &
Equipment**

Equity

**Insurance
liabilities**

Provisions



Caramels – Insurance companies

Capital Adequacy
Asset Quality
Reinsurance
Actuarial
Management
Earnings (Profitability)
Liquidity & Funding
Sensitivity to Market Risks



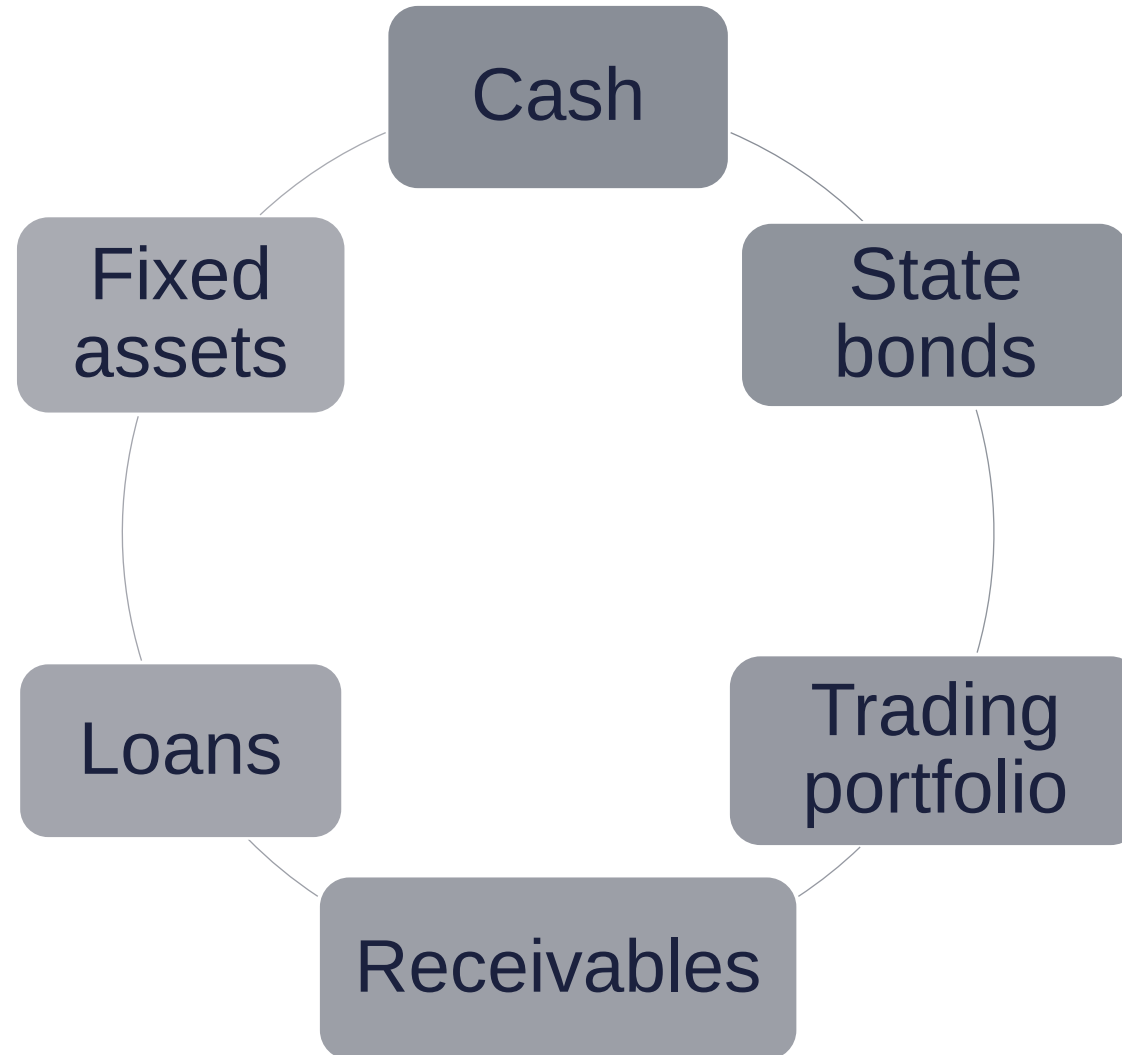
Minimum is subject to local regulation.

Extra reserve reflects more conservative strategy.

$$\frac{\textit{Capital}}{\textit{Total Assets}}$$

$$\frac{\textit{Capital}}{\textit{Total (or Net) claims}}$$

Asset quality



Significant for non-life insurance business.

Reinsurance's share of premiums and claims may be very different in each period.

$$\textit{retention ratio} = \frac{\textit{Net Premiums}}{\textit{Gross premiums}}$$

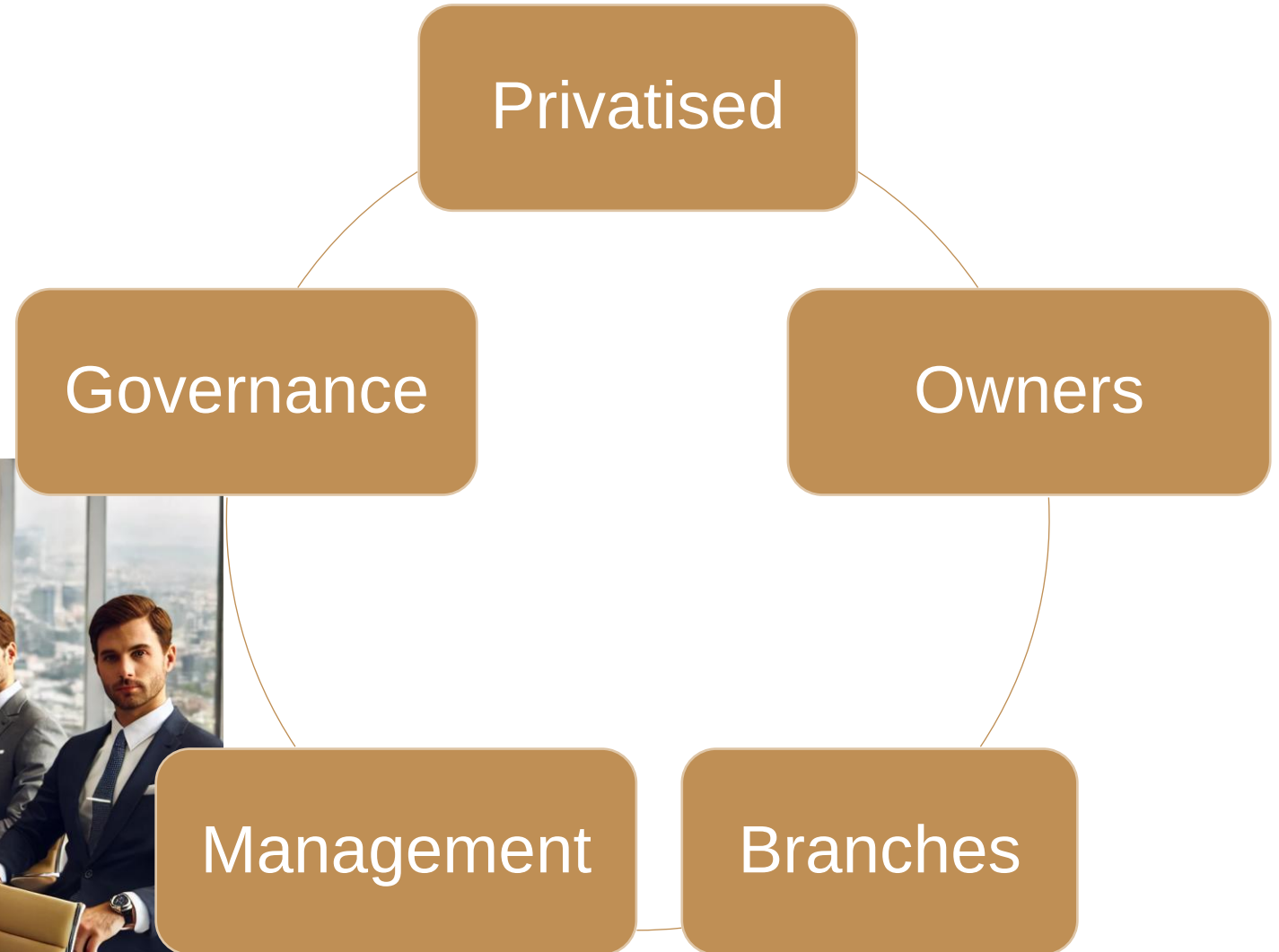
What insurance risk remains with the insurance firm?

Net technical reserves

Average Net Claims over the last 3 years

Net technical reserves

Average Net Premiums over the last 3 years



Earnings & Profitability

Strategic
focus

Macro
situation

Regional
structure

Interest vs
non-interest
income

Growth
trends

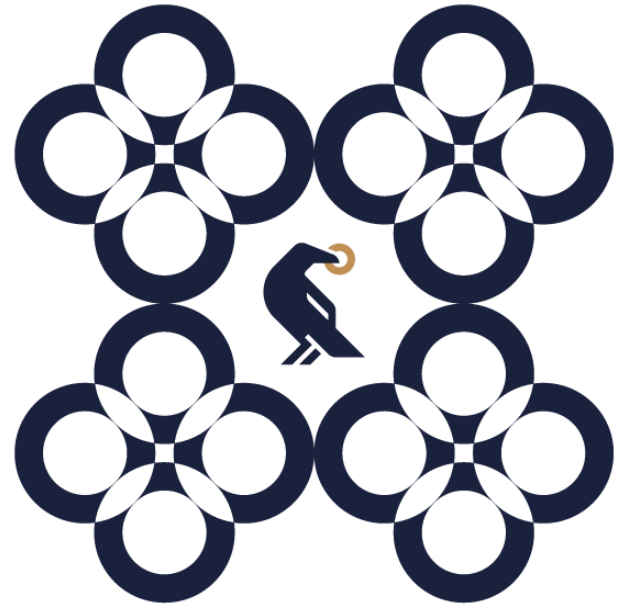
Extraordinary
events

How to measure profitability?

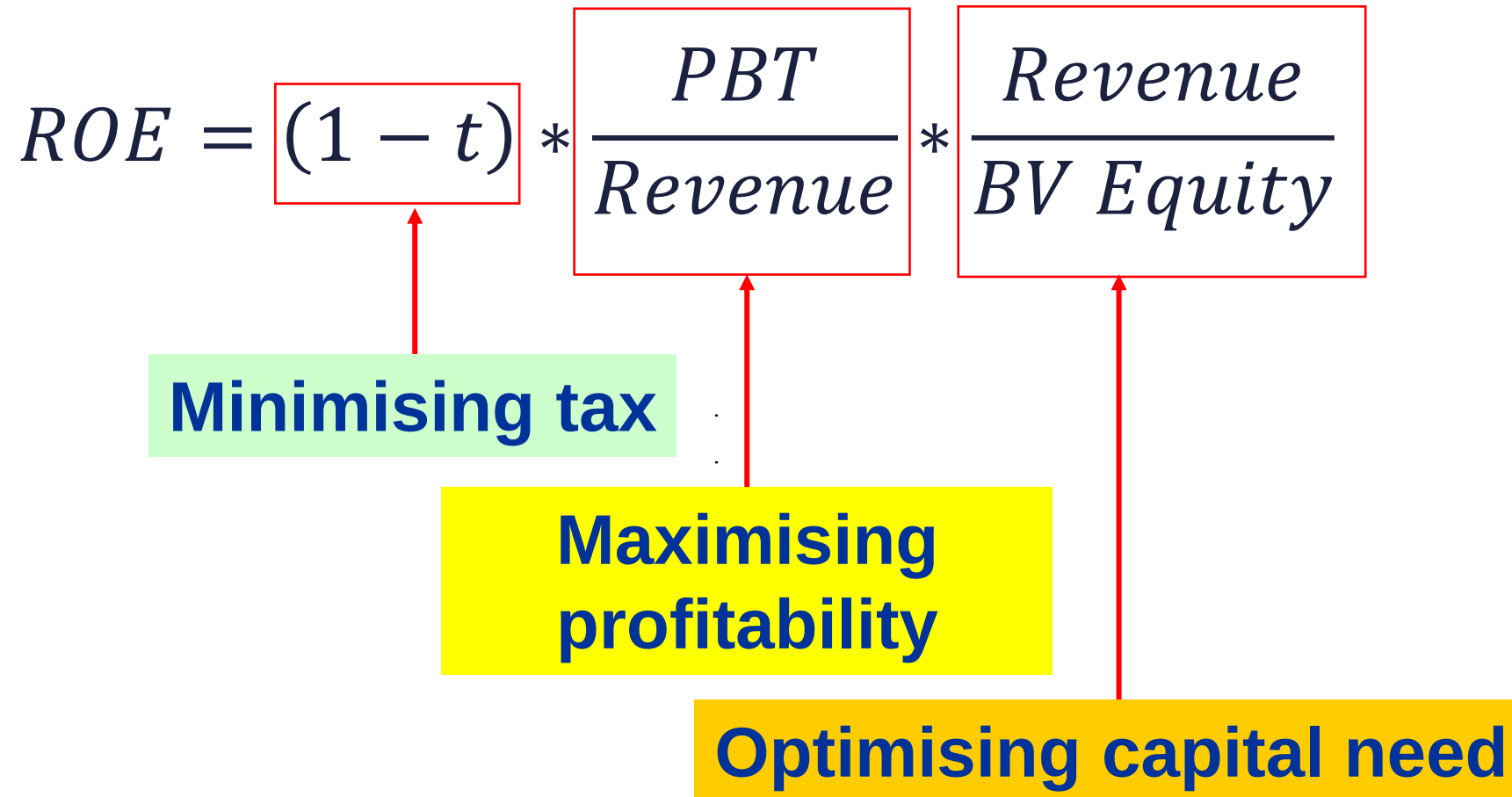
$$\begin{aligned} \text{Combined ratio} &= \\ &= \frac{\text{Gross Claims}}{\text{Gross premiums}} + \frac{\text{Operating expenses}}{\text{Gross premiums}} \end{aligned}$$

$$\begin{aligned} \text{Technical ratio} &= \\ &= \frac{\text{Expected loss}}{\text{Gross premiums}} + \frac{\text{Commissions}}{\text{Gross premiums}} \end{aligned}$$

Valuation issues



The strategic view of ROE



DCF valuation methods

Method	To discount	Discount factor	Assessment
DDM	Dividend	Required equity return (r_E)	For easy to predict dividends (announced policy) on a mature market.
Direct equity DCF valuation	FCFE	r_E	Detailed hectic financing plan, bottom-up CF.
Economic profit	RI	r_E	Shows when and how much value is created.

Leading back multiples to the fundamentals

$$P_0 = \frac{Div_1}{r_E - g}$$

$$\begin{aligned}
 & \rightarrow P_0 / E_1 = \frac{(1 - b)}{r_e - g} & b = \frac{\text{Dividend}}{\text{EPS}} \\
 & \rightarrow P_0 / BV_0 = \frac{ROE_1 * (1 - b)}{r_e - g} \\
 & \rightarrow P_0 / S_1 = \frac{\text{Net margin} * (1 - b)}{r_e - g}
 \end{aligned}$$

where

$$\text{Net margin} = \frac{\text{Profit after tax}}{\text{Sales}}$$

Sector-specific multiples

P/DIV

P/tBV

P/Deposits

Multiples' Summary Ranking (1990-2018)

		EU Investment Banks	EU Large Commercial Banks	EU Small Commercial Banks	US Investment Banks	US Large Commercial Banks	US Small Commercial Banks
Best choice	1	P/E (FY2)	P/E (FY2)	P/E (FY1)	P/E (FY2)	P/E (FY2)	P/E (FY2)
	2	P/E (FY1)	P/E (FY1)	P/E (FY2)	P/E (FY1)	P/E (FY1)	P/E (FY1)
	3	P/E (LTM Basic no Extra)	P/E (LTM Basic no Extra)	P/E (LTM Basic no Extra)	P/Common Dividends	P/E (LTM Diluted no Extra)	P/BV
	4	P/BV	P/Common Dividends	P/Common Dividends	P/E (LTM Diluted no Extra)	P/BV	P/TBV
	5	P/TBV	P/Deposits	P/Revenues	P/Deposits	P/TBV	P/E (LTM Diluted no Extra)
	6	P/Common Dividends	P/Revenues	P/Deposits	P/Revenues	P/Deposits	P/Revenues
	7	P/Deposits	P/BV	P/BV	P/BV	P/Revenues	P/Deposits
Worst choice	8	P/Revenues	P/TBV	P/TBV	P/TBV	P/Common Dividends	P/Common Dividends

<https://www.fondazioneoiv.it/wp-content/uploads/2018/06/bank-valuation-using-multiples-in-us-and-europe.pdf>

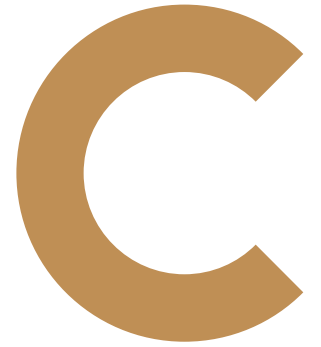
Industry multiples in Europe, 2023 June

Financials Sector/Industry Group	Market Cap/Revenues	P/TBV	P/E	P/B
Financials	2.0x	1.3x	9.4x	1.1x
Banks	2.1x	0.9x	6.8x	0.8x
Financial Services ¹	2.4x	2.3x	12.5x	1.5x
Insurance	0.7x	2.0x	11.3x	1.3x

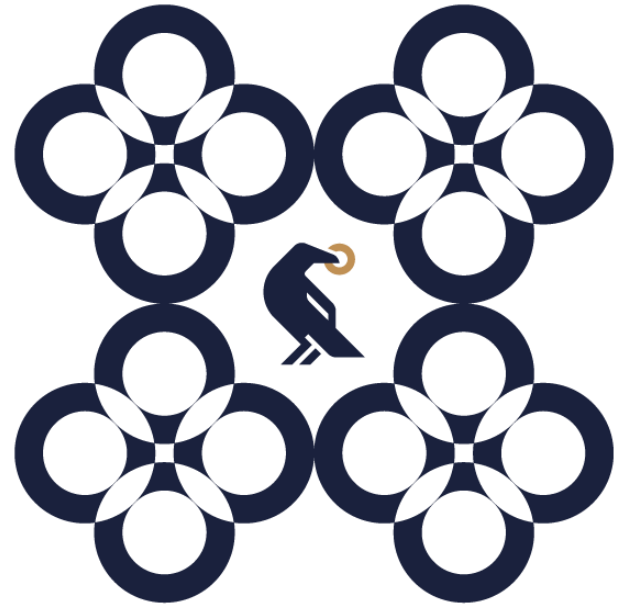
¹ Reflects revisions to the GICS structure that took effect on March 17, 2023. Comparisons to prior periods may not be meaningful due to the shifting composition of the structure. For more information, see <https://www.msci.com/our-solutions/indexes/gics>.

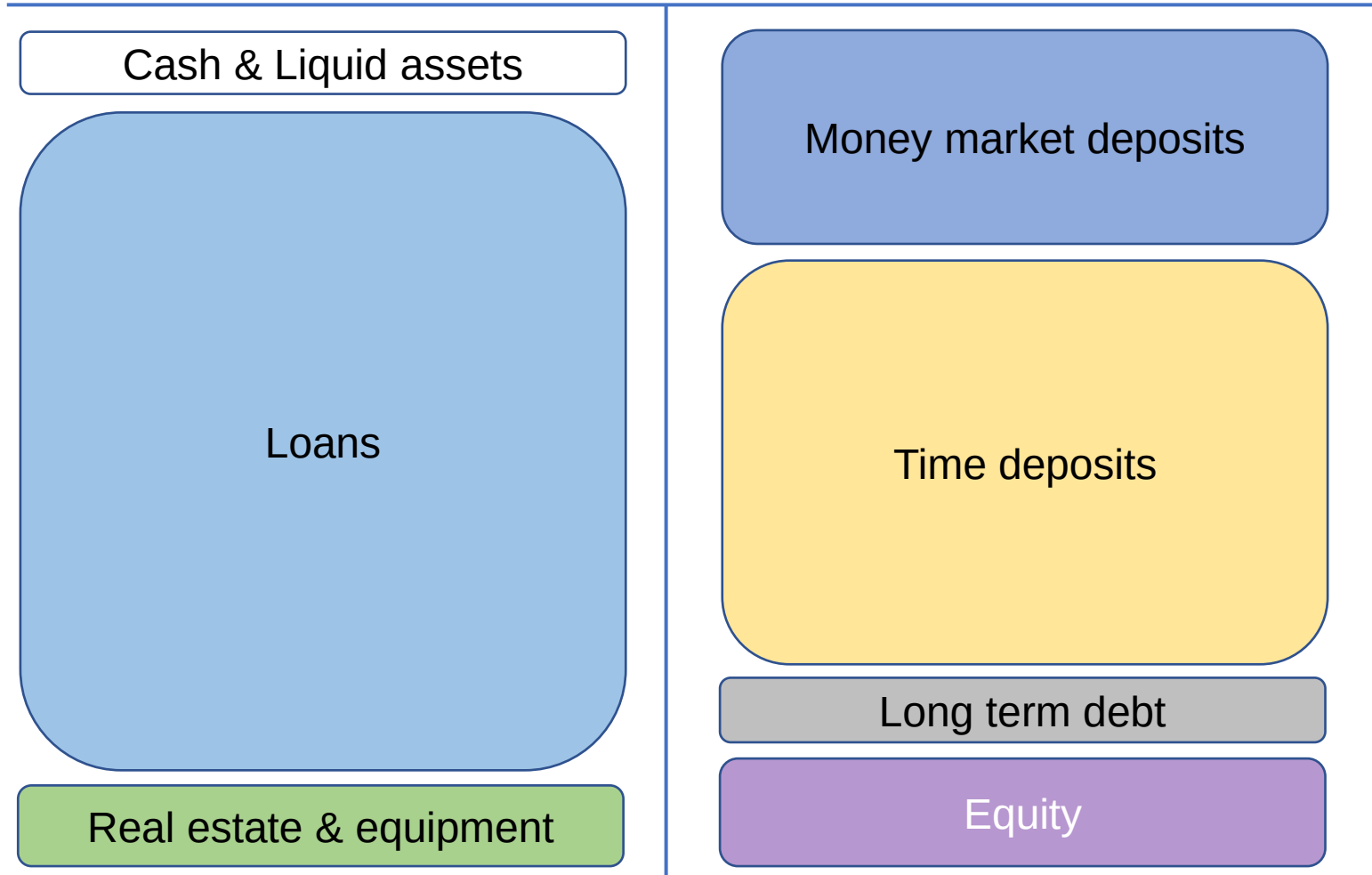
Source: https://media-cdn.kroll.com/jssmedia/kroll-images/pdfs/industry-multiples-europe-q2-2023-executive-summary.pdf?_ga=2.37749347.1784869111.1733654450-1193548733.1733654450

**Thank you
for your attention.
Any questions?**

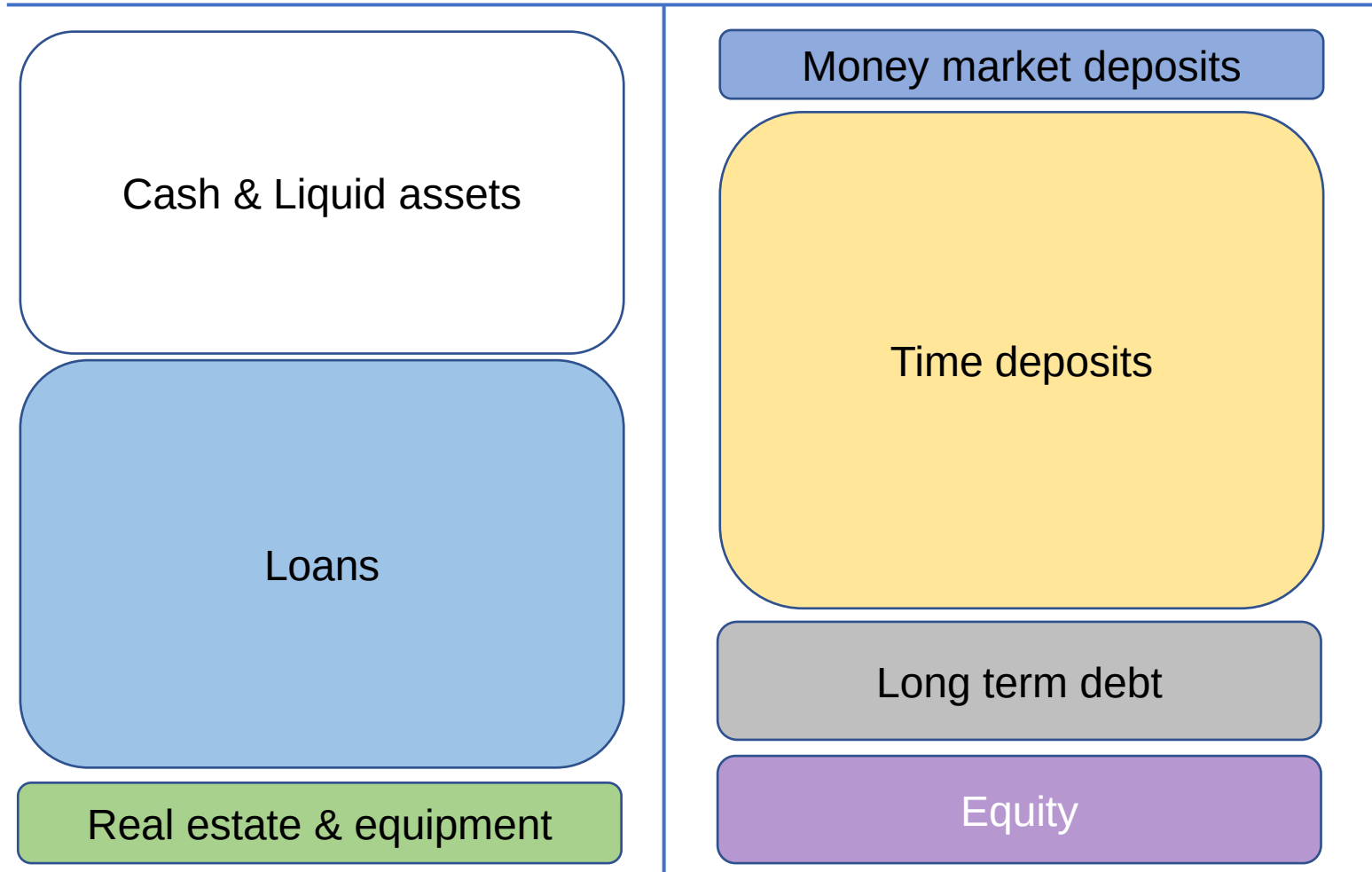


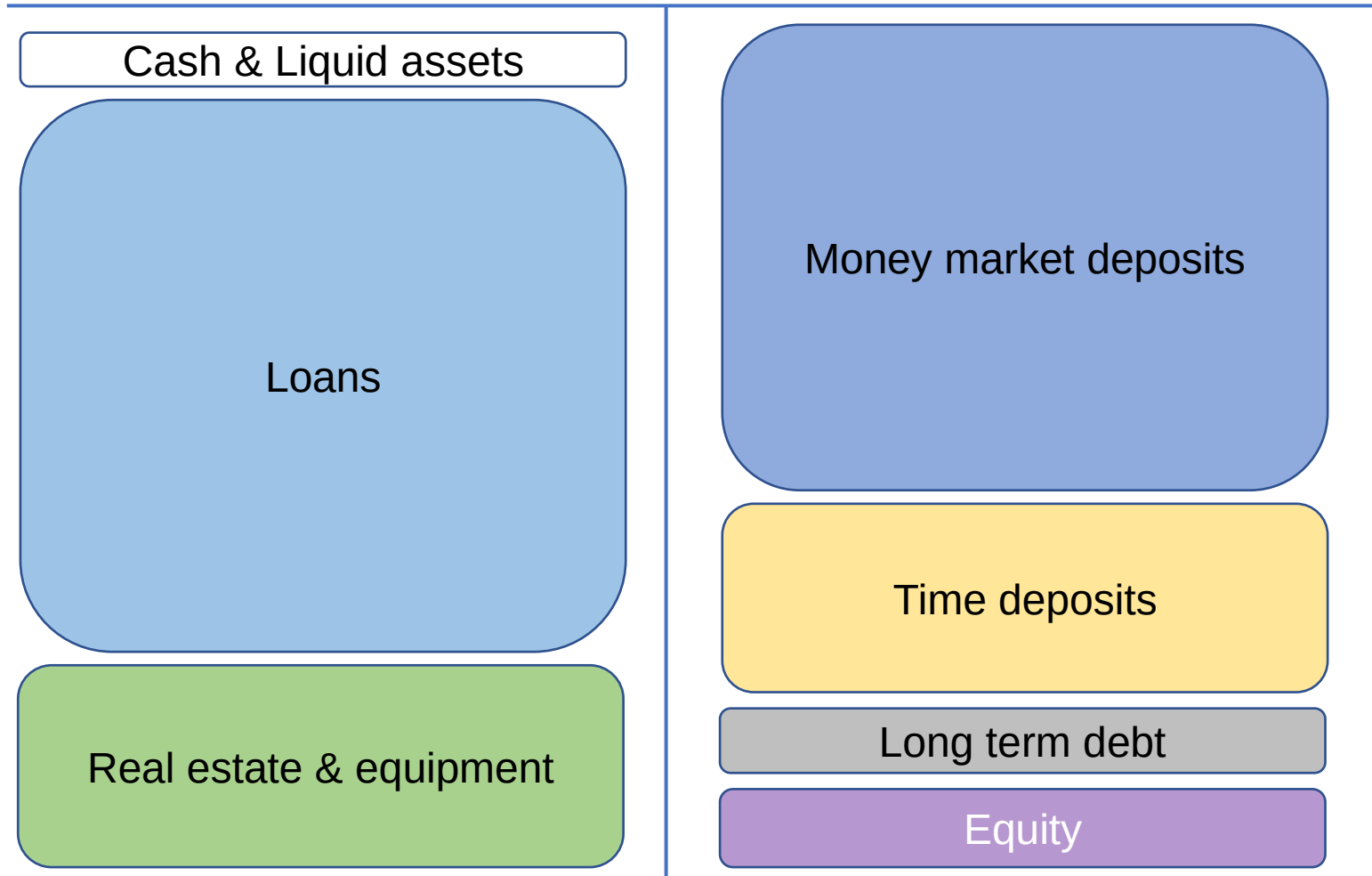
Reserve slides for Banks

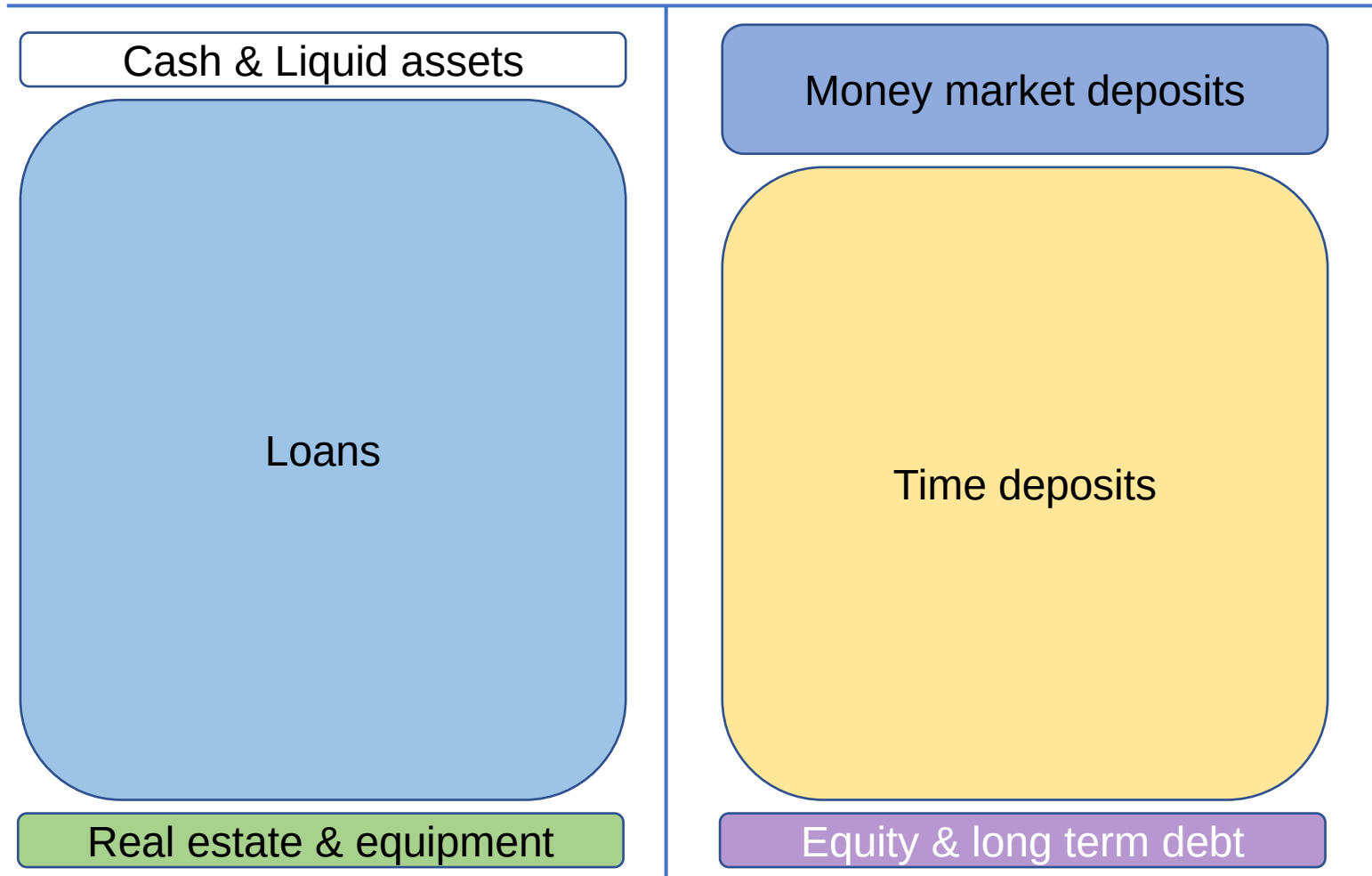




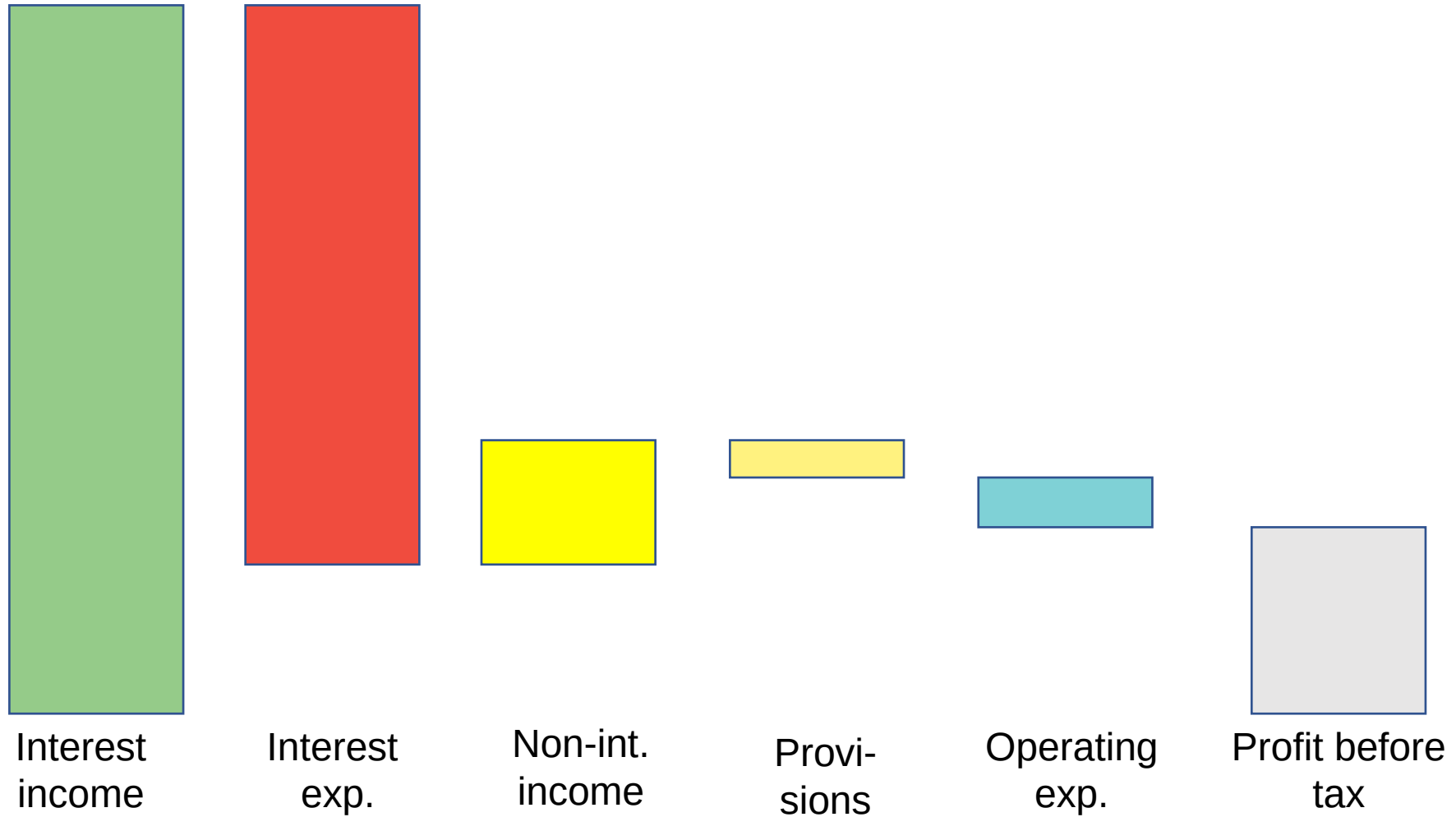
Low profitability





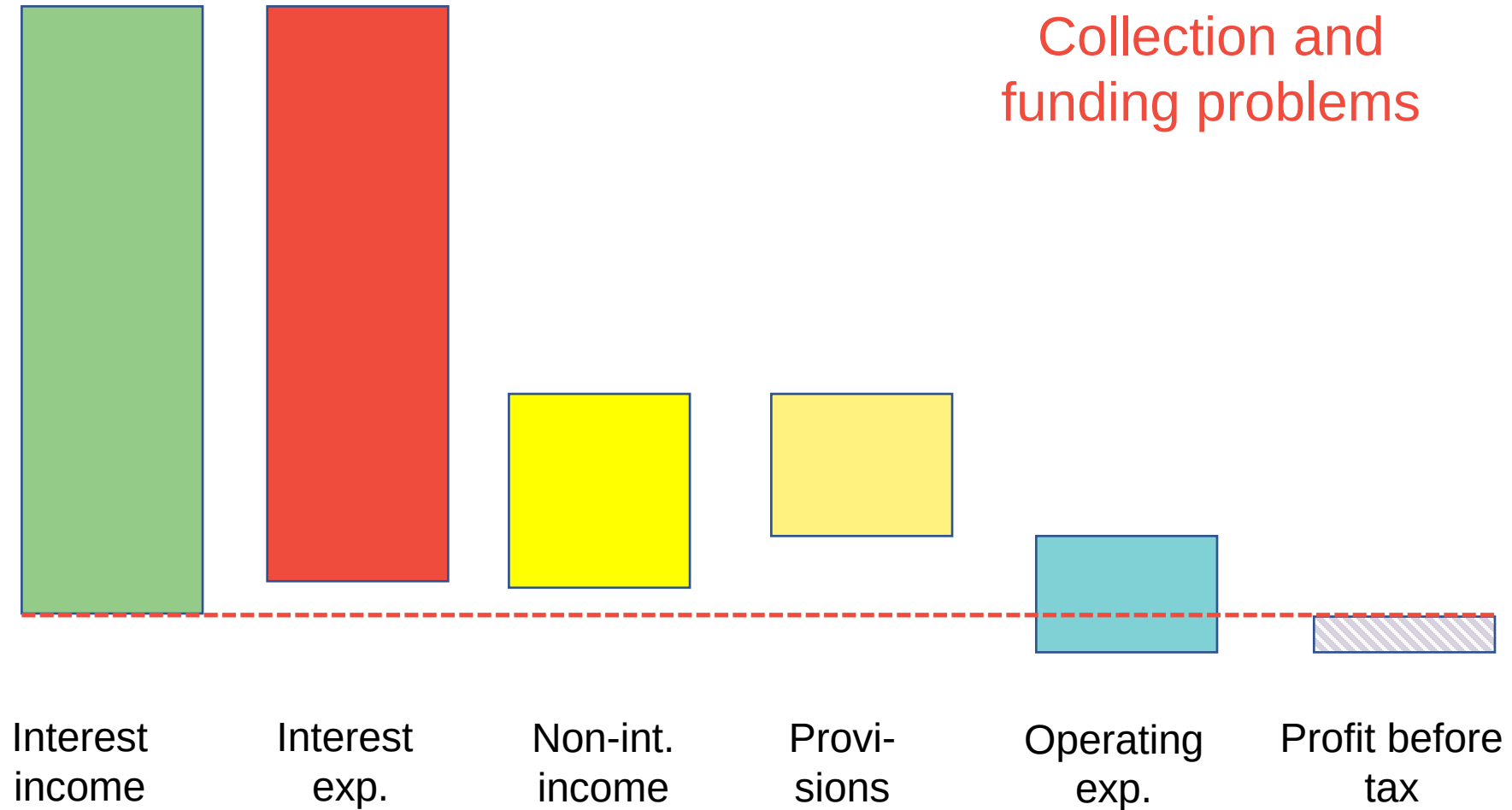


Base case



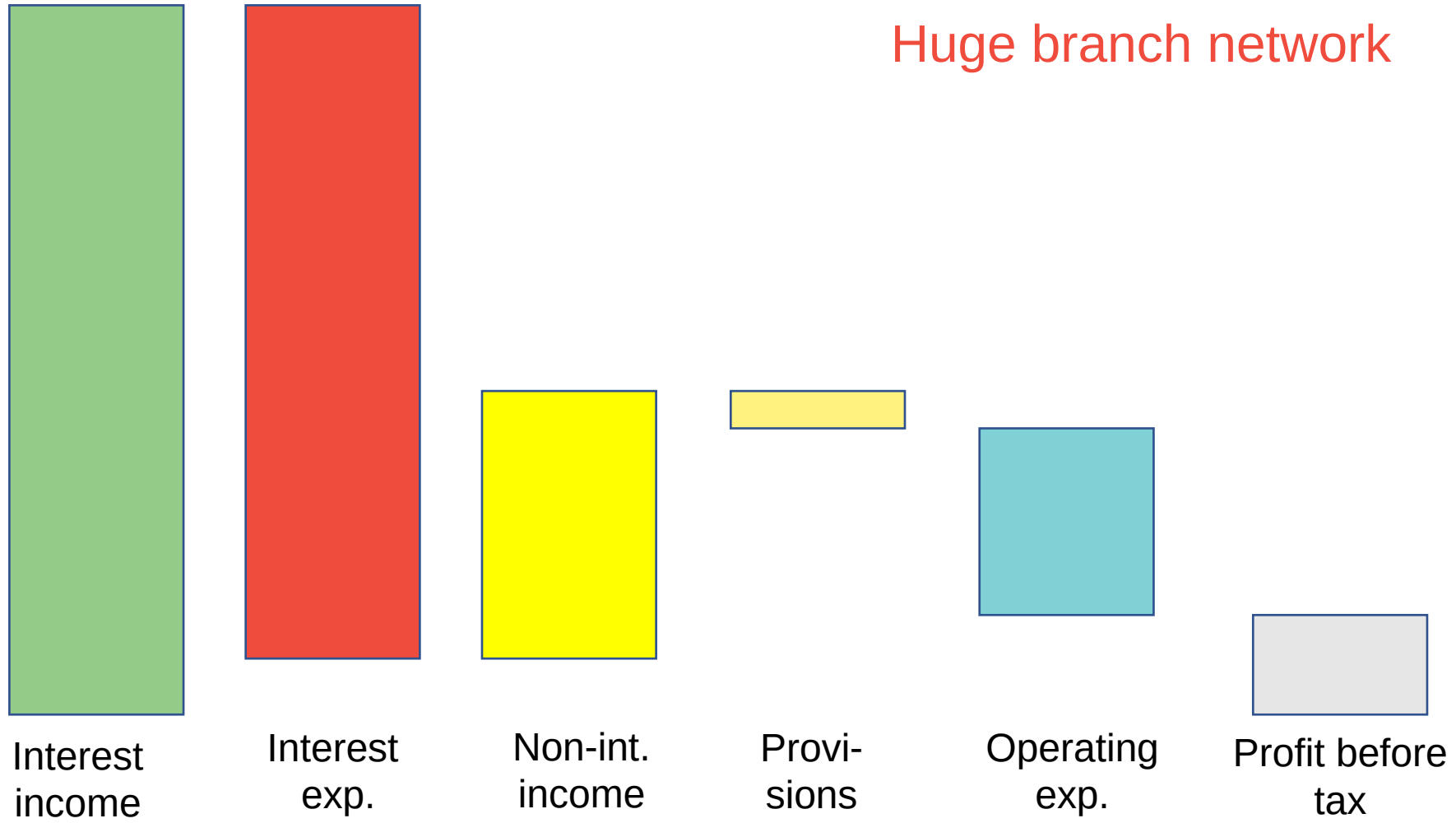
What is the issue? 1

Collection and
funding problems



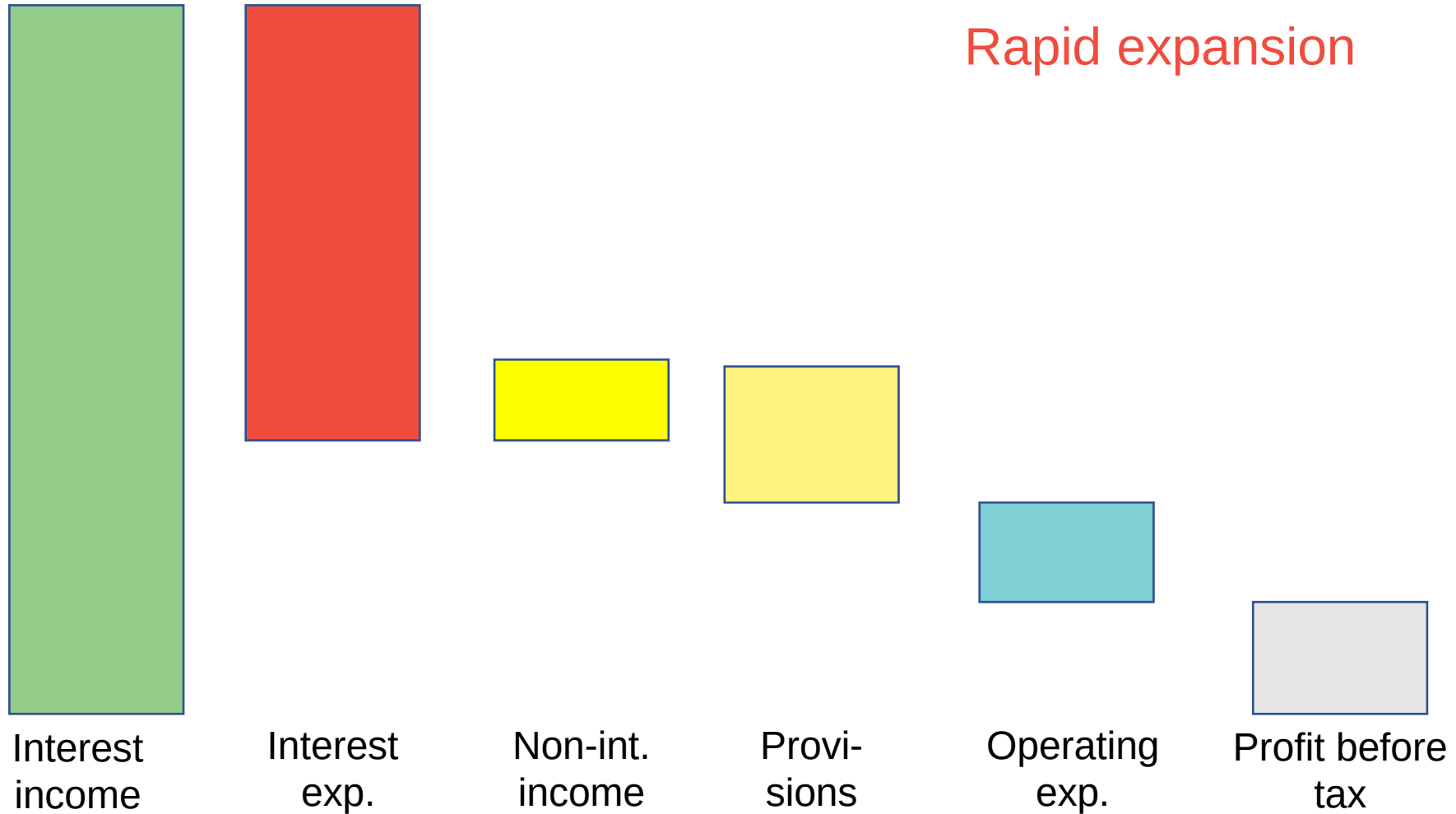
What is the issue? 2

Huge branch network

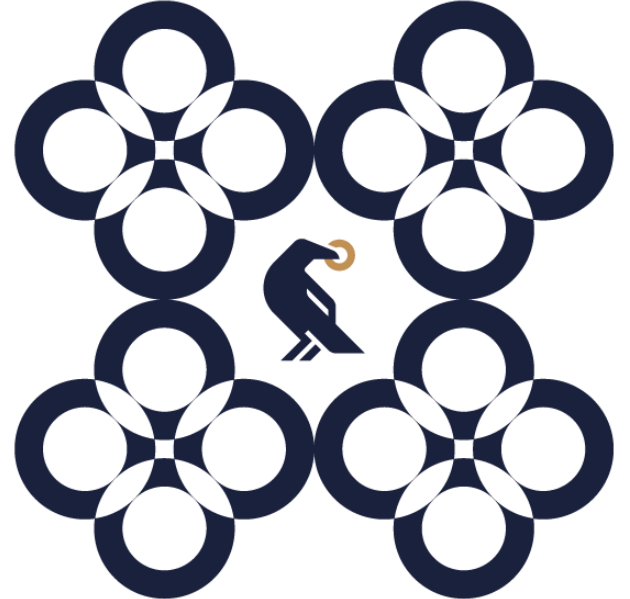


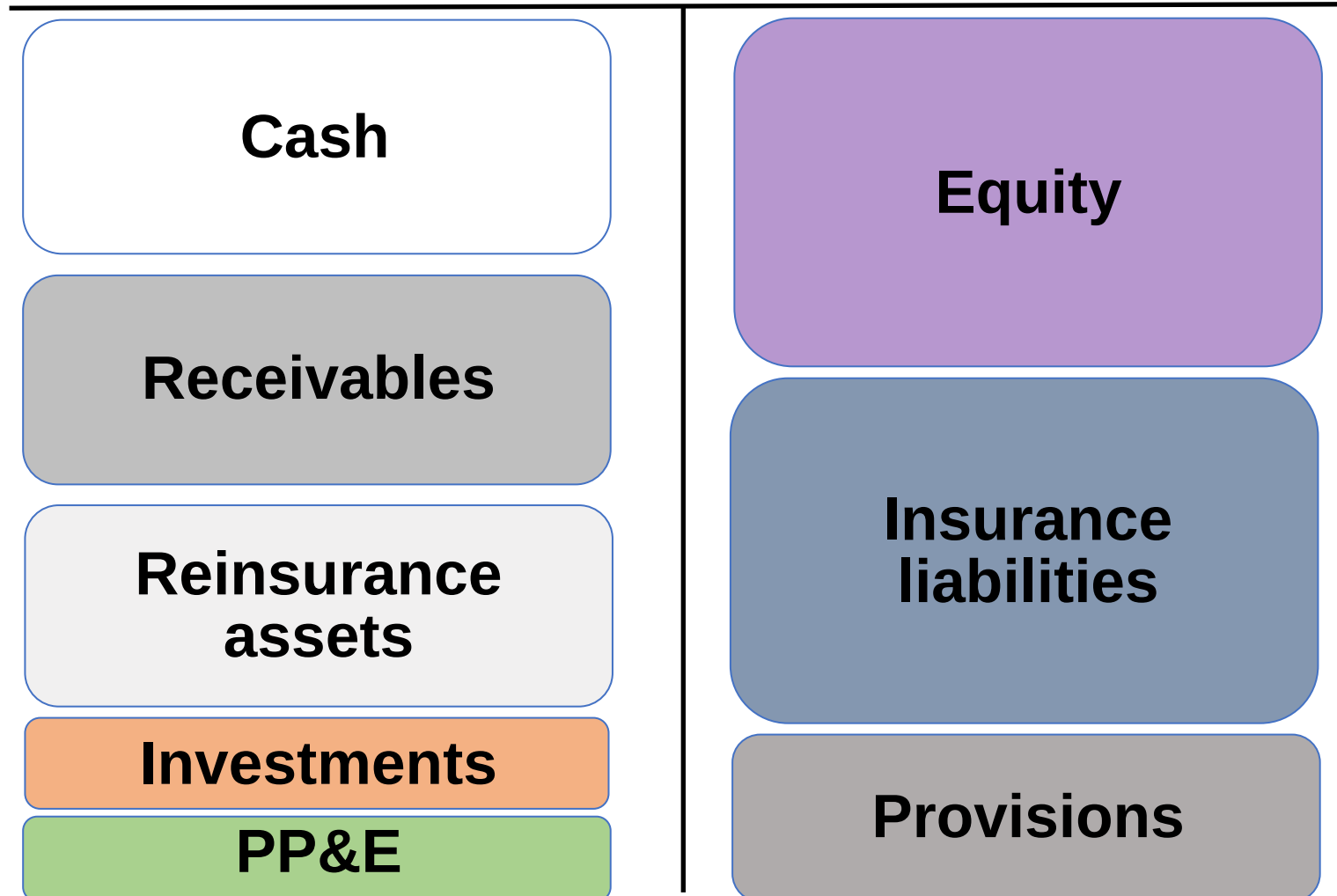
What is the issue? 3

Rapid expansion

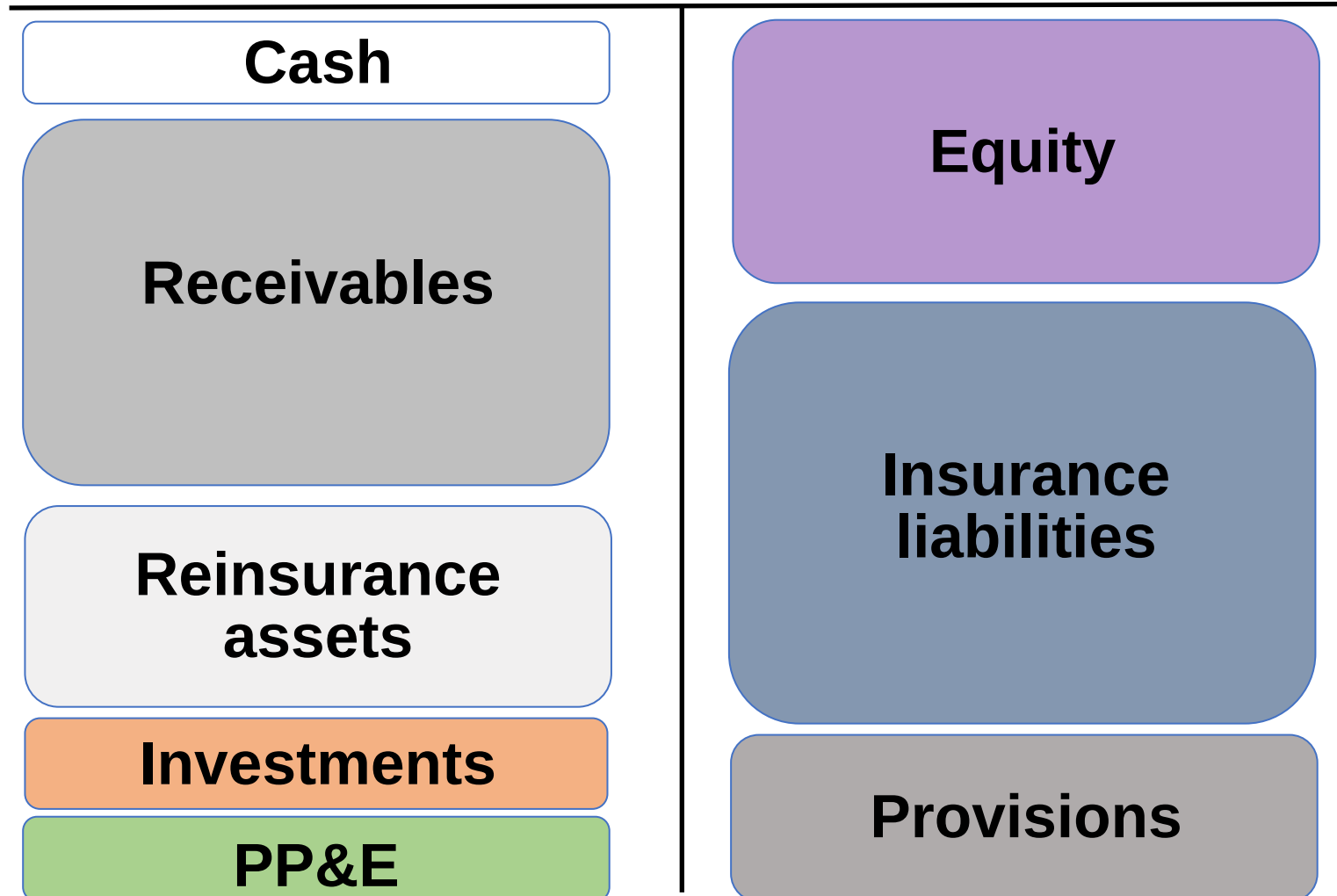


Reserve slides for Insurance Companies

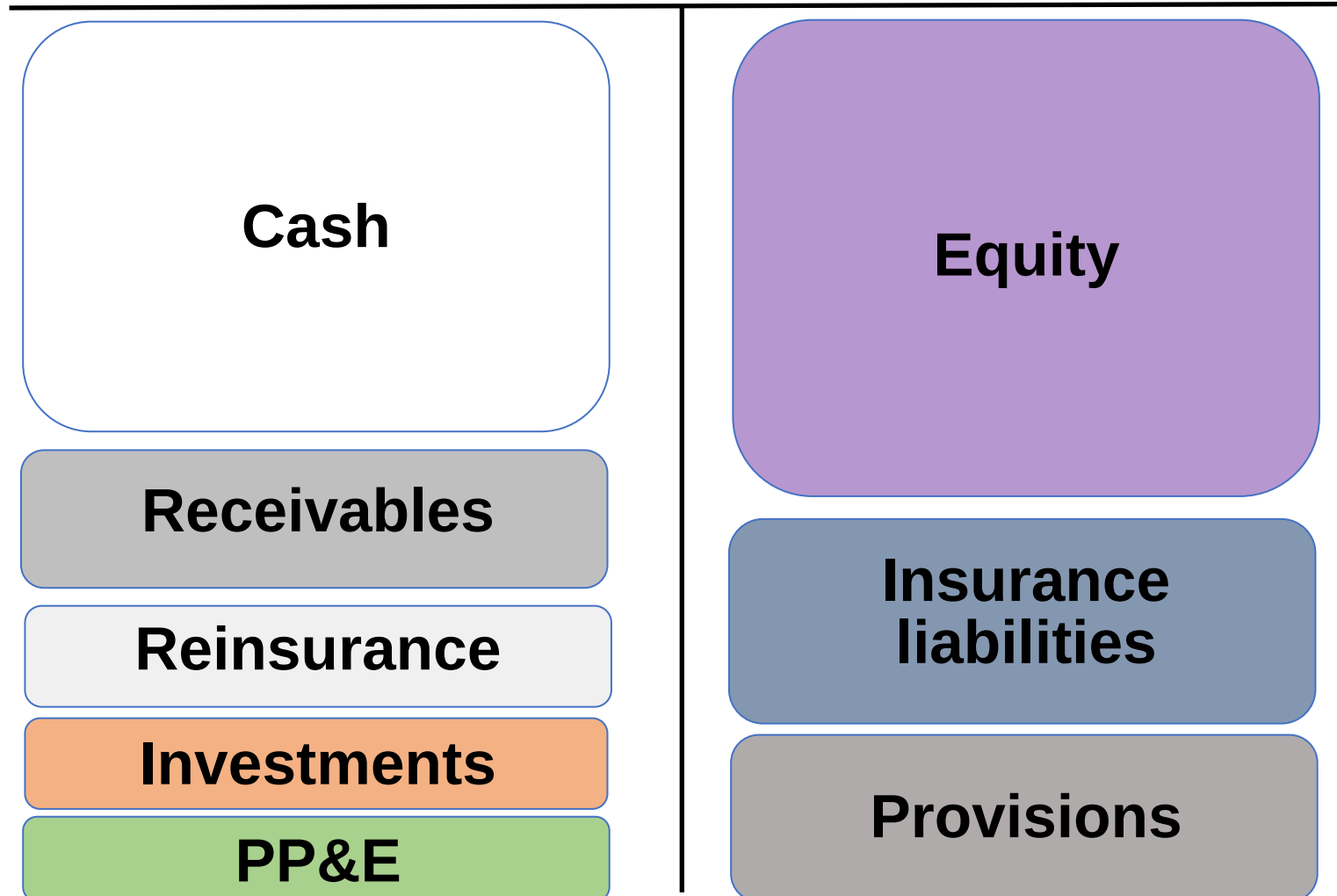


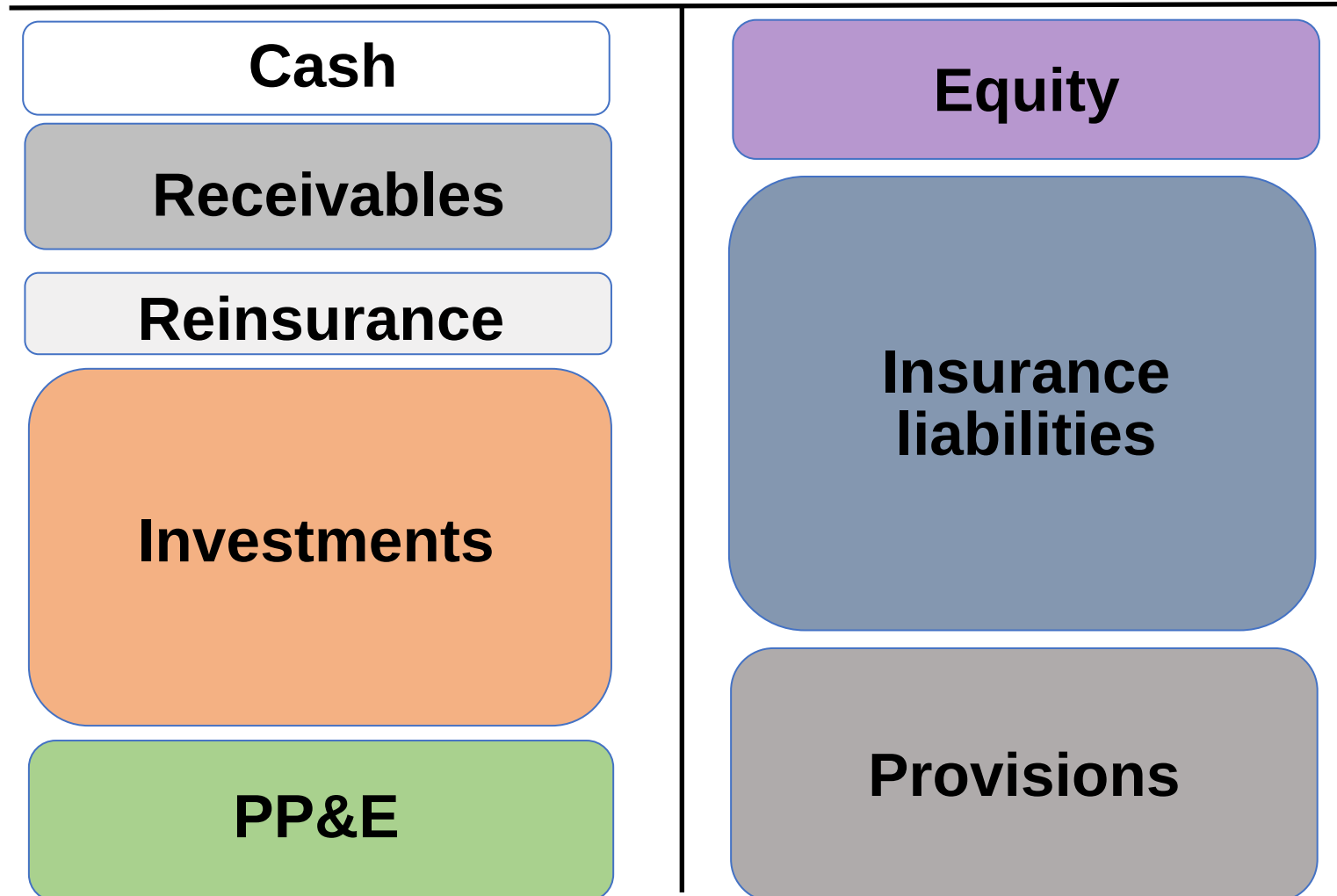


Liquidity problems



No operation?

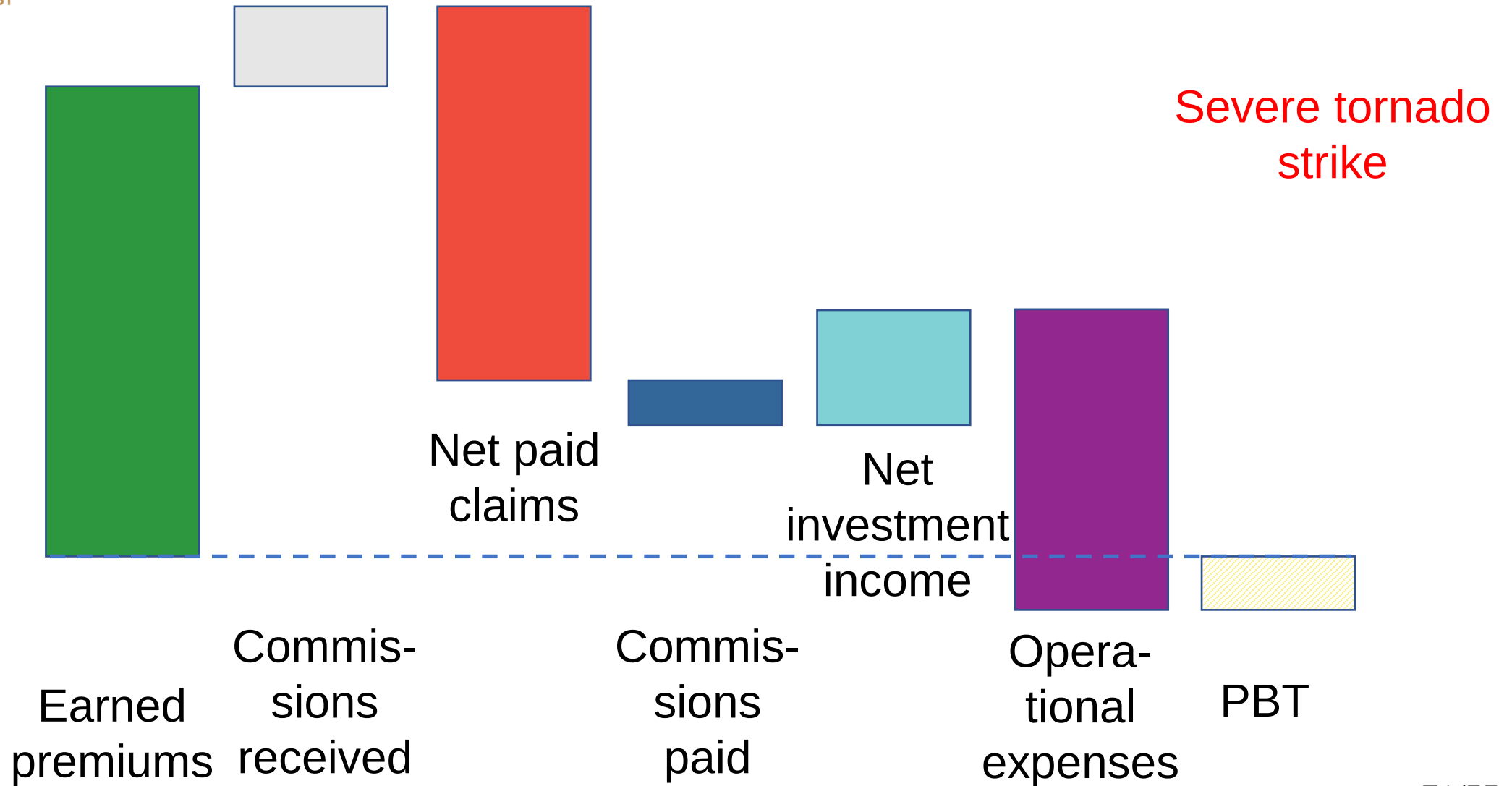




After good years

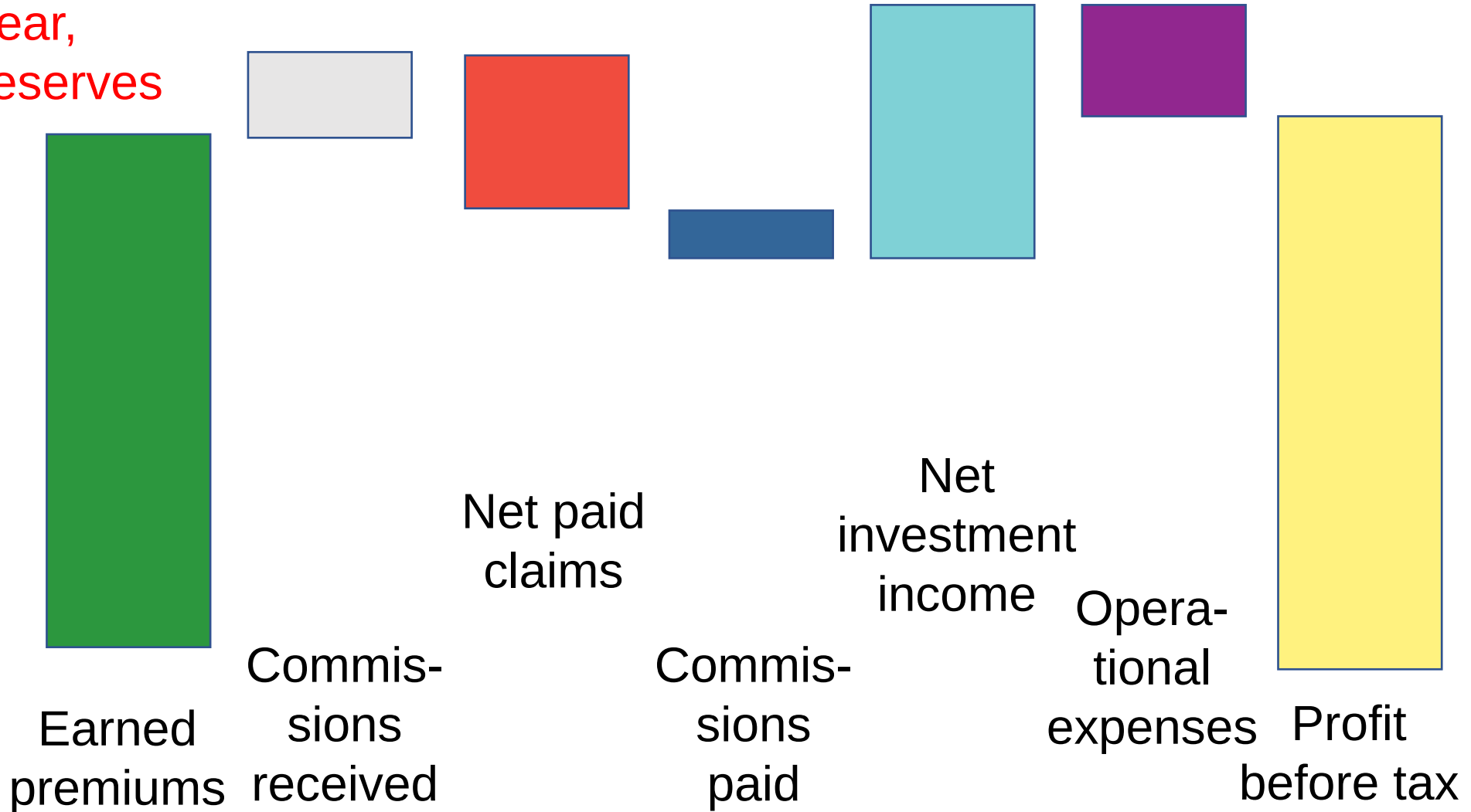
Cash	Equity
Receivables	Insurance liabilities
Reinsurance assets	Provisions
Investments	
PP&E	

What happened? 2

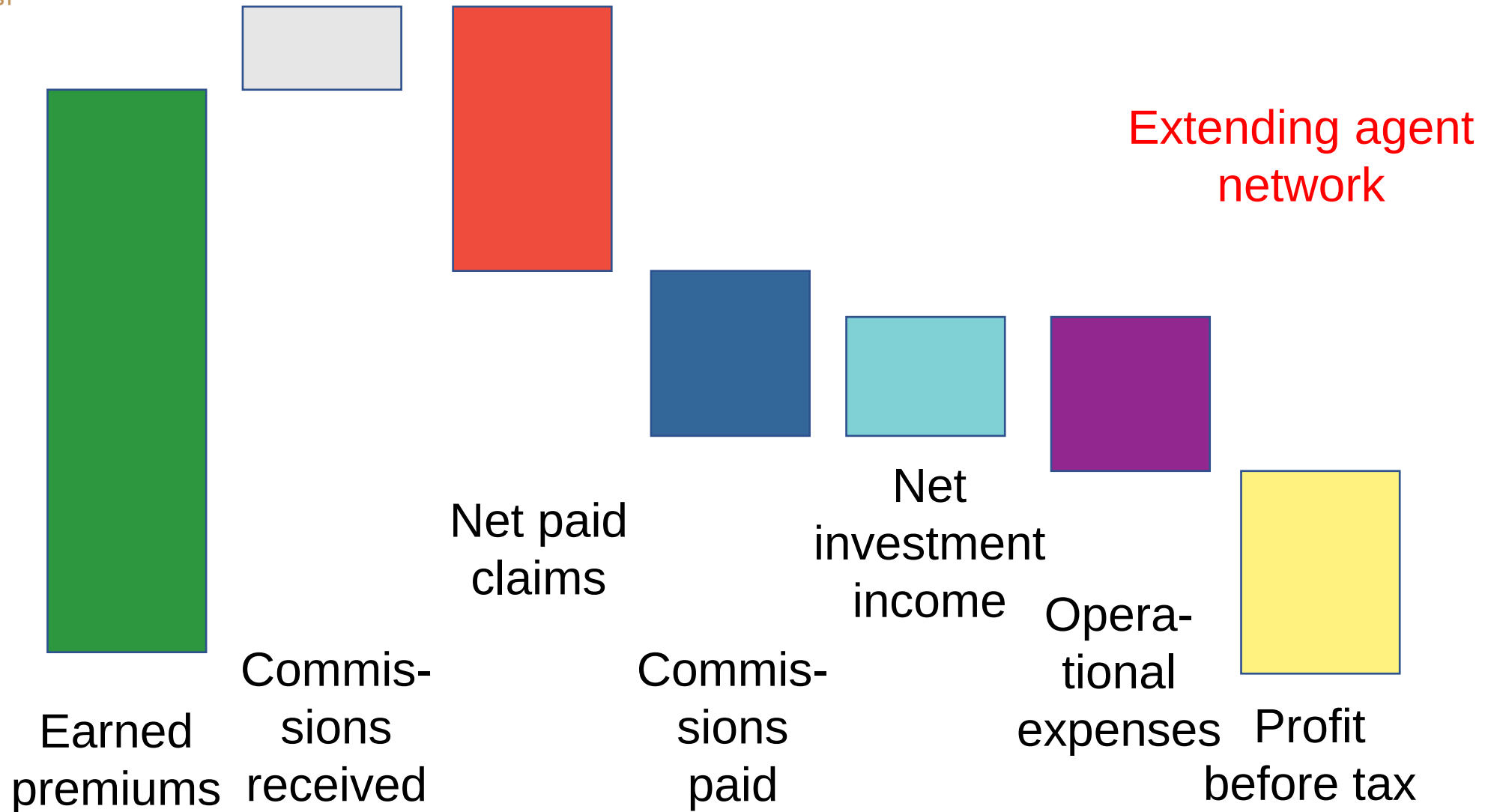


What happened? 3

Quiet year,
collecting reserves

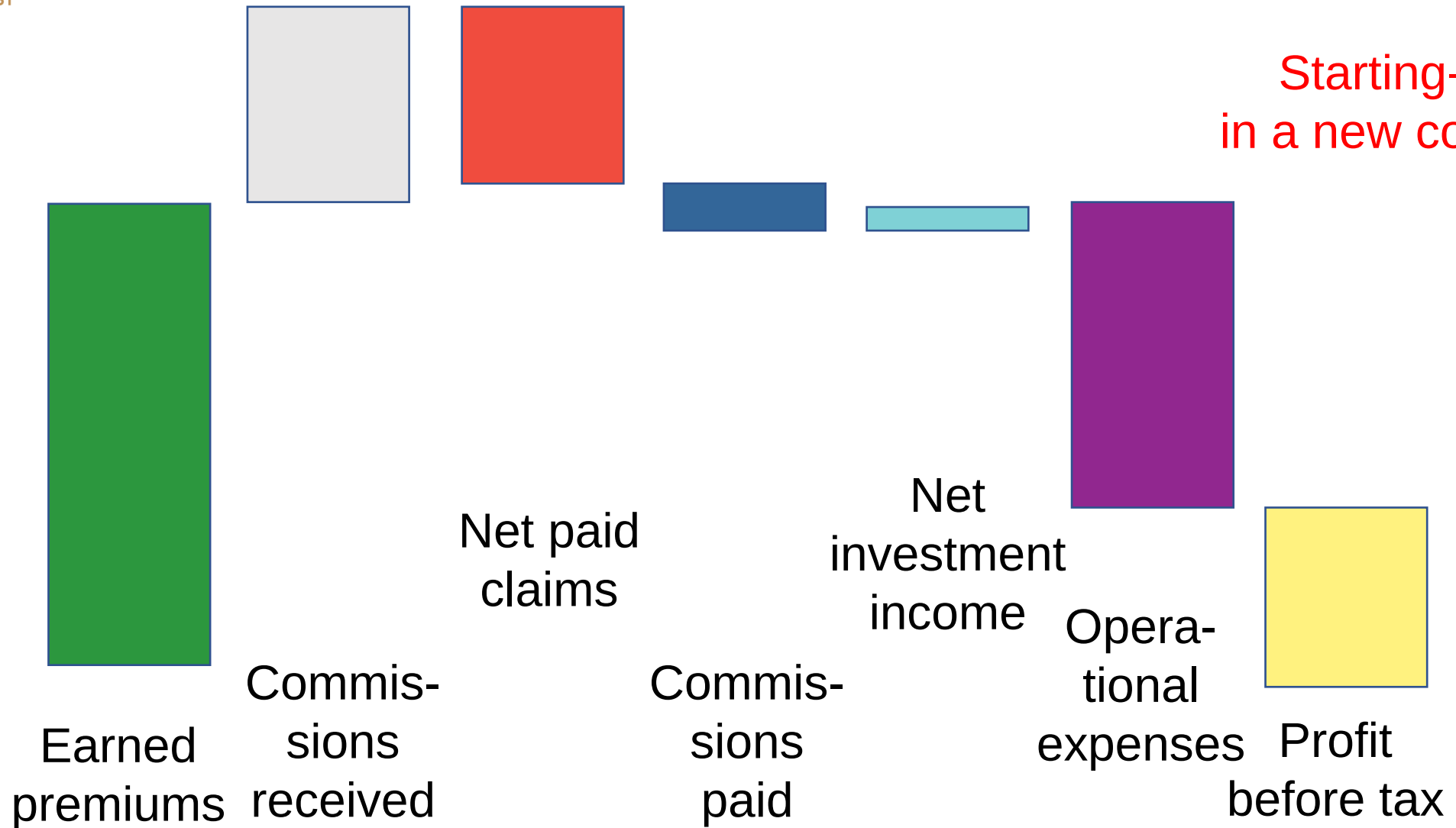


What happened? 4



What happened? 5

Starting-up
in a new country



What happened? 6

